MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MICROPROCESSOR, DIGITAL, KEY CODE HRH

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic microprocessor module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements . Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 1.540 failures per million operating hours for class I or 17.232 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be HRH.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/237-1 for class I and M28787/237-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, V_{BB} = -5.0 ±0.5 V dc, and V_{DD} = 12 ±1.2 V dc except that the voltage limits of table IV apply only for power supply stress check.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

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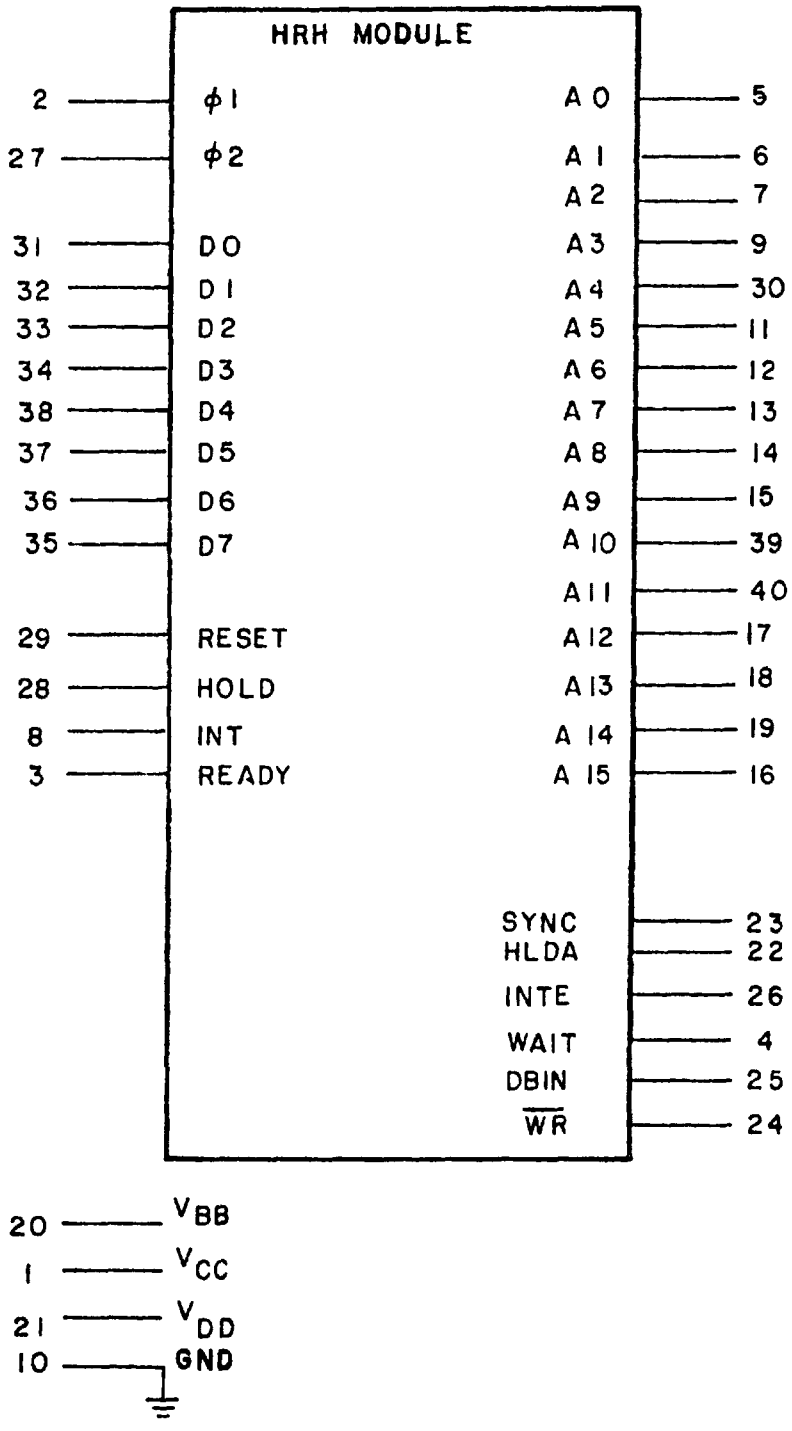


FIGURE 1. HRH module function diagram, microprocessor.

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* TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B5	Power supply stress check	IV	S	← Must pass all remaining tests →										
C1-C30	Output voltage, low level	V	X	0	450	0	450	0	450	0	450	0	450	mV
D1-D29	Output voltage, high level	VI	X	3.7	4.5	3.7	4.5	3.7	4.5	3.7	4.5	3.7	4.5	V
E1	Power supply current (V_{BB})	XII	X	0	1	0	1	0	1	0	1	0	1	mA
E2	Power supply current (V_{CC})	XII	X	10	100	10	100	10	100	10	100	10	100	mA
E3	Power supply current (V_{DD})	XII	X	10	80	10	80	10	80	10	80	10	80	mA
F1-F6	Input current, high level	XIII	X	-10	10	-10	10	-10	10	-10	10	-10	10	µA
F7-F14	Input current, high level	XIII	X	-100	100	-100	100	-100	100	-100	100	-100	100	µA
G1-G8	Input current, data bus, 3.7 V level	XIV	X	-0.01	-2.00	-0.01	-2.00	-0.01	-2.00	-0.01	-2.00	-0.01	-2.00	mA
H1-H6	Input current, low level	XV	X	-10	10	-10	10	-10	10	-10	10	-10	10	µA
H7-H14	Input current, low level	XV	X	0	-100	0	-100	0	-100	0	-100	0	-100	µA
J1-J24	Three-state output leakage current, low level	XVII	X	0	-100	0	-100	0	-100	0	-100	0	-100	µA
K1-K24	Three-state output leakage current, high level	XVIII	X	0	10	0	10	0	10	0	10	0	10	µA

See footnotes at end of table.

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* TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
L1-L4	Dynamic functional	XX	X	← See table XX →										
M1-M4	Dynamic functional and propagation delay time	XXV	Q	← See table XXV →										
N1-N30	Output transition time (t_{TLH})	XXXI	Q	3	35	3	35	3	35	3	35	3	35	ns
P1-P30	Output transition time (t_{THL})	XXXII	Q	2	25	2	25	2	25	2	25	2	25	ns
Q1-Q3	Power supply transient amplitude	XXXIII	Q	0	400	0	400	0	400	0	400	0	400	mV
R1-R3	Power supply filter capacitance	XXXIV	S	4.06	7.54	4.06	7.54	4.06	7.54	4.06	7.54	4.06	7.54	µF
S1,S2	Input capacitance	XXXV	Q	2	25	2	25	2	25	2	25	2	25	pF
S3-S14	Input capacitance	XXXV	Q	2	15	2	15	2	15	2	15	2	15	pF
T1-T22	Output capacitance	XXXV	Q	2	20	2	20	2	20	2	20	2	20	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. If test conditions A or B are part of a sequence of tests, the order of testing shall be as follows: A, B where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions C through R shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 7. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition R, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3. Following the life test and while still at the maximum class temperature, electrical tests C through R shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests C through T shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

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6.4 Module function. The module is a complete 8-bit parallel, fixed instruction set, central processing unit (CPU) for use in general purpose digital computer systems. The block diagram of the module is shown on figure 2. The major functional features of the CPU are as follows:

a. Six 8-bit general purpose registers and an accumulator.

b. General purpose registers can be addressed individually or in pairs for both single and double precision.

c. Four testable flags for arithmetic and logical instructions.

d. Fifth flag for decimal arithmetic.

e. External stack stores accumulator, flags, program counter, and six general purpose registers.

f. 16-address lines.

g. 8-line bidirectional data bus.

h. Separate control lines.

i. Direct memory access (DMA) operation.

j. Interrupt operation.

k. Reset capability.

6.4.1 Input and output relationships. The operations performed by the CPU are controlled primarily by the instruction (op code) presented to the data bus during an instruction fetch cycle and by signals applied to the control inputs (RESET, HOLD, INTERUPT, and READY).

6.4.1.1 Functional description of pins.

Function designation	Function description
A15-A0	ADDRESS BUS. Provides unique address for memory and I/O devices. The bus is capable of addressing 64k 8-bit words and 256 output devices. A0 is the least significant bit. The bus is three-state output.
D7-D0	DATA BUS. Provides 8-bit bidirectional communications between the CPU, memory, and I/O devices for instruction and data transfers. The 8-bits outputed during the first clock cycle of a machine cycle describes the status of the CPU. The bus is three-state bidirectional and D0 is the least significant bit.
SYNC	This output provides an indication of the beginning of each machine cycle.
DBIN	This output indicates to other circuits that the data bus is in its input mode. This signal should be used to enable the gating of data from memory or I/O onto the data bus.
READY	This input indicates to the microprocessor that valid memory or input data is available on the data bus. This signal is used to synchronize the microprocessor with slow I/O or memory. If after

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Function designation	Function description
	addressing a device, READY is not returned, then the microprocessor will enter a wait state for as long as READY is low.
WAIT	A high level on the WAIT output indicates that the CPU is in a WAIT state.
$\overline{WR}$	This output is used for memory or I/O write control. The data on the data bus is stable when the $\overline{WR}$ signal is low.
HOLD	The HOLD input requests the microprocessor to enter the HOLD state. The HOLD state allows an external device to gain control of the address and data bus as soon as the microprocessor has completed its use of these buses for the current machine cycles. It is recognized under the following conditions: • The microprocessor is in the HALT state. • The microprocessor is in state T2 or TW state (WAIT) and the READY signal is active. As a result of entering the HOLD state the ADDRESS BUS (A15-A0) and DATA BUS (D7-D0) will be in their high-impedance state. The microprocessor acknowledges the HOLD state with the HLDA pin.
HLDA	This output signal appears in response to the HOLD request and indicates that the address and data bus lines will go to their high-impedance mode.
INTE	The interrupt enable output indicates the state of the internal interrupt enable flip/flop.
INT	The interrupt request input signal causes the CPU to enter its interrupt routine at the end of its current instruction unless it is in its HOLD state or its interrupt enable flip/flop is reset.
RESET	The RESET IN input signal clears the program counter and resets the INTE and HLDA flip/flops. After this signal, the program will start at memory location 0. The flags, accumulator, stack pointer, and registers are not cleared.
01, 02	The clock inputs are two phase, nonoverlapping clocks. They are not TTL compatible.
V_{SS}	Ground reference.
V_{DD}	+12 V ±10 percent.
V_{CC}	+5 V ±10 percent.
V_{BB}	-5 V ±10 percent.

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6.4.1.2 Machine cycle. With the exception of a DAD instruction, the basic machine cycle consists of those operations necessary to reference an addressable memory or I/O device and process the information. Note that the machine cycle is composed of some number of states where a state is defined as the interval between two successive positive going transitions of the Ø1 clock. The five types of machine cycles are as follows and the operations normally associated with each are as follows:

T1	A memory address or I/O device number is placed on the address bus and the cycle type is specified as defined in 6.4.1.3 by information on the data bus.
T2	The CPU checks the READY and HOLD inputs and also checks for the halt instruction.
TW (optional)	When the READY line is low the CPU enters the wait states until the ready line returns high. If the microprocessor executes a halt instruction, it will enter the wait state and remain there until a RESET, HOLD, or INT (with interupt enabled) occurs.
T3	Data is transferred to or from memory or I/O device during this cycle. Note that this data is an instruction if the machine cycle is an instruction fetch.
T4, T5 (optional)	These states are used by the CPU for instructions which require additional states for internal data movement of ALU operations.

6.4.1.3 Status words. There shall be 10 machine cycle types referred to as: instruction fetch, memory read, memory write, stack read, stack write, input read, output write, interupt acknowledge, halt acknowledge, and interupt acknowledge while halt.

6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 603) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and may be obtained upon request.

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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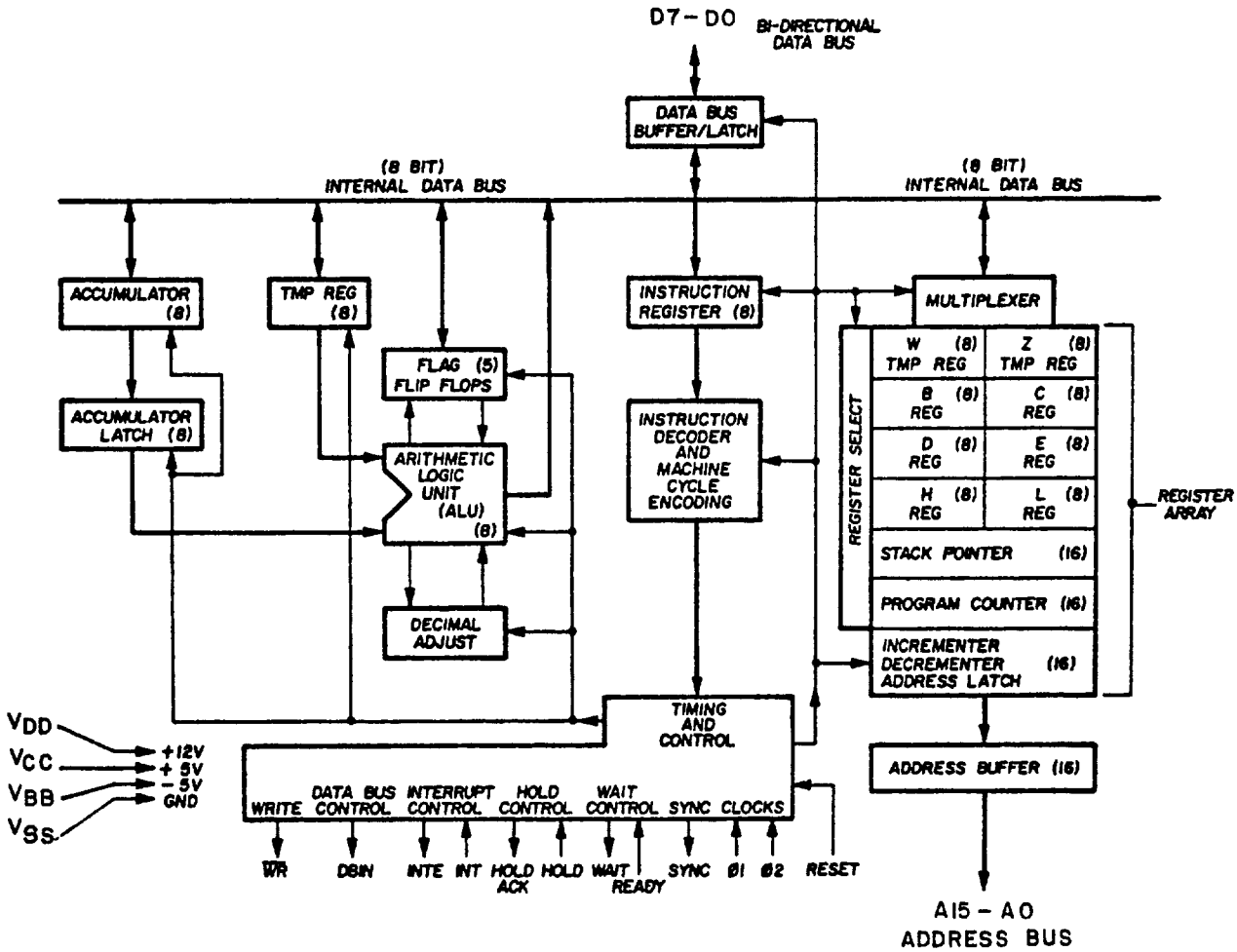
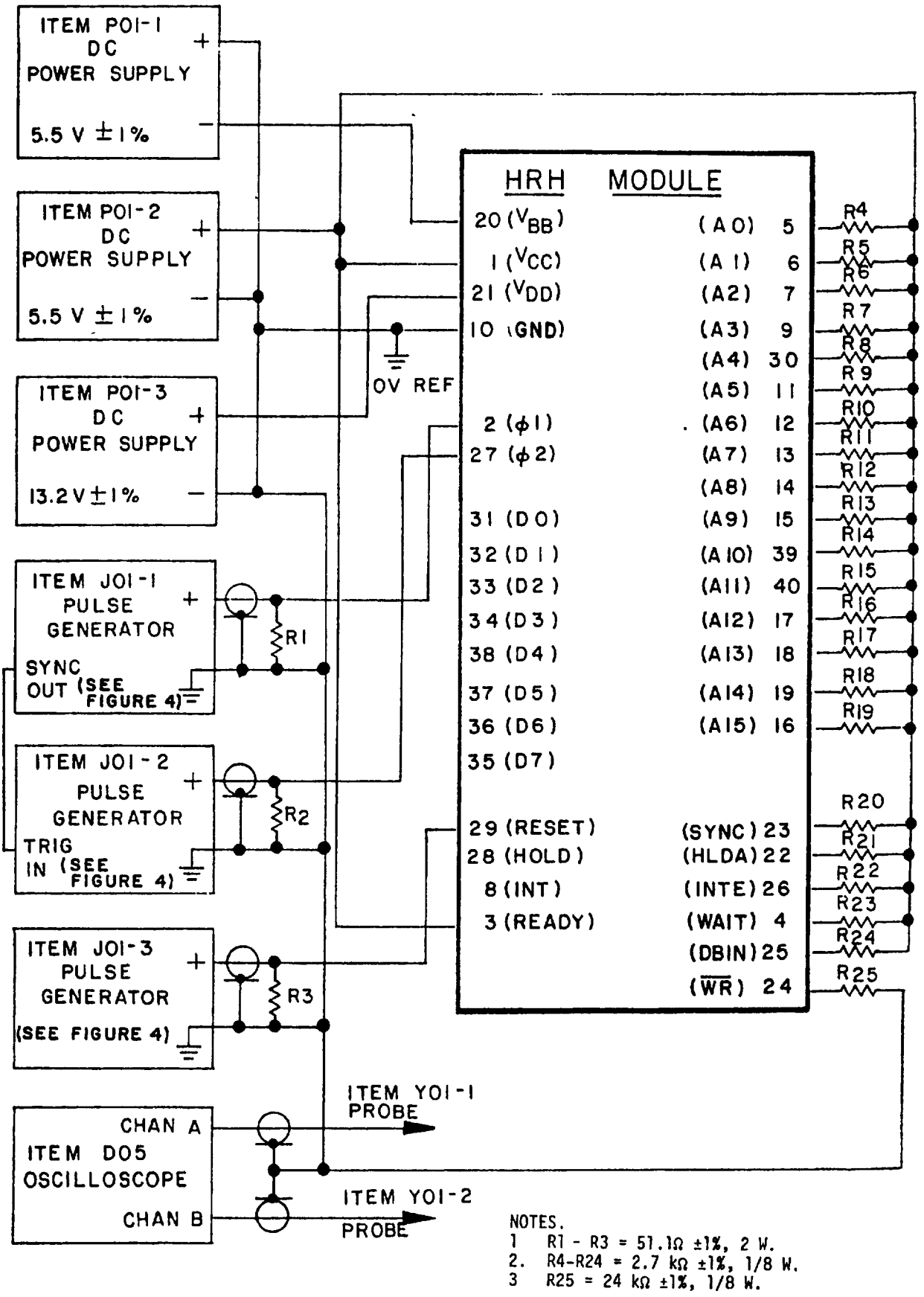


FIGURE 2. Functional block diagram.

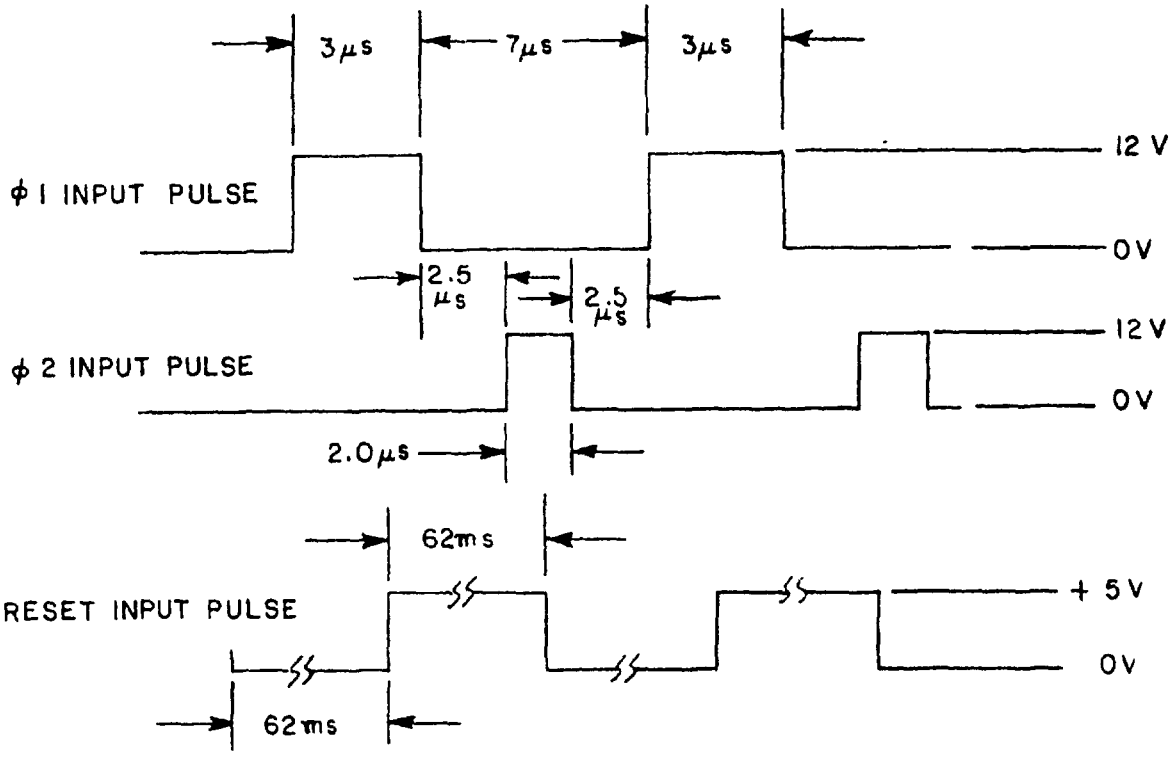
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NOTES.
1 R1 - R3 = 51.1Ω ±1%, 2 W.
2. R4-R24 = 2.7 kΩ ±1%, 1/8 W.
3 R25 = 24 kΩ ±1%, 1/8 W.

FIGURE 3. Operating temperature and life test circuit.

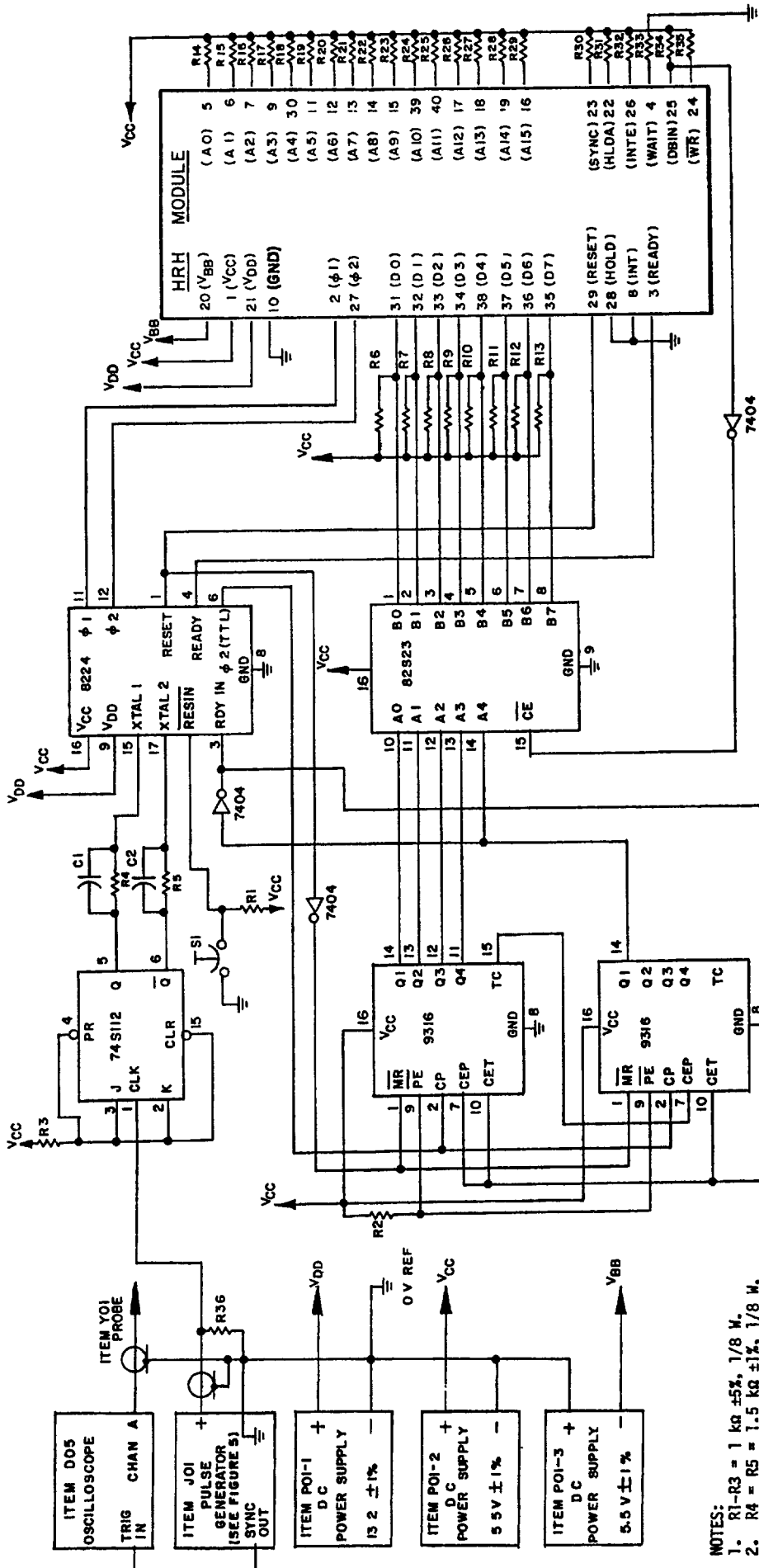
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NOTES:
1. Input pulse parameters:
   Ø1: $t_{TLH} = t_{THL}$ = 450 ns maximum.
   Ø2: $t_{TLH} = t_{THL}$ = 750 ns maximum.
   RESET: t_{TLH} = 500 ns maximum.
2. Waveform transition times (t_{TLH} and t_{THL}) are measured from 1 to 8 V levels for clocks and from 0.5 to 4.5 V levels for the RESET input. All measurements are with the device under test connected.

FIGURE 4. Operating temperature and life test circuit waveforms.

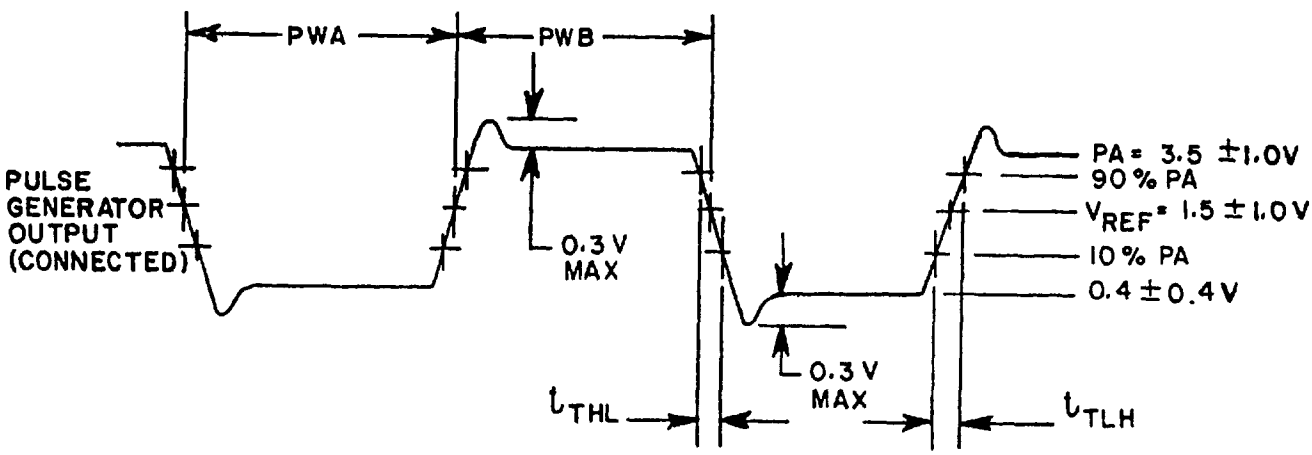
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NOTES:
1. R1-R3 = 1 kΩ ±5%, 1/8 W.
2. R4 = R5 = 1.5 kΩ ±1%, 1/8 W.
3. R6-R32, R34 = R35 = 2.7 kΩ ±1%, 1/8 W.
4. R33 = 24 kΩ ±1%, 1/8 W.
5. R36 = 51.1Ω ±1%, 2 W.
6. C1 = C2 = 150 pF ±10%.
7. Intergrated circuits: 74S112, 82S23, 7404, 8224, and 9316.
8. S1 = SPST normally open momentary contact push button switch.
9. Adjust item J01 to 18 MHz ±1% square wave with pulse amplitude of 3.5 ±1.0 V; then push switch S1 and verify that pins 31-34, 38, 37, 36, 35 (D0-D7), 5, 6, 7, 9, 30, 11, 12, 13, 14, 15, 39, 40, 17, 18, 19, 16 (A0-A15), 23 (SYNC), 22 (HLDA), 26 (INTE), 25 (DBIN), 24 ($\overline{WR}$), are less than 0.45 V maximum and that pin 4 (WAIT) voltage is greater than 3.7 V minimum.
10. See table II for PROM pattern.

FIGURE 5. Component temperature test circuit.

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NOTE: Item J01 output pulse parameters·

PWA = 55 ±5 ns.
PWB = 55 ±5 ns.
$t_{TLH} = t_{THL}$ = 30 ±20 ns.

FIGURE 6. Component temperature test circuit waveform.

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TABLE II. Component temperature test circuit PROM pattern.

Address	OUTPUT B0	B1	B2	B3	B4	B5	B6	B7	MNEMONIC
0	L	L	L	L	L	L	L	L	
1	L	L	L	L	L	L	L	L	
2	L	L	L	L	L	L	L	L	
3	L	H	H	H	H	H	L	L	MV1 A, L
4	L	L	L	L	L	L	L	L	
5	L	L	L	L	L	L	L	L	
6	L	L	L	L	L	L	L	L	
7	L	L	L	L	L	L	L	L	
8	L	L	L	L	L	L	L	L	
9	L	L	L	L	L	L	L	L	
10	H	H	L	L	H	L	H	H	OUT, L
11	L	L	L	L	L	L	L	L	
"	"	"	"	"	"	"	"	"	
"	"	"	"	"	"	"	"	"	
"	"	"	"	"	"	"	"	"	
"	"	"	"	"	"	"	"	"	
31	"	"	"	"	"	"	"	"	

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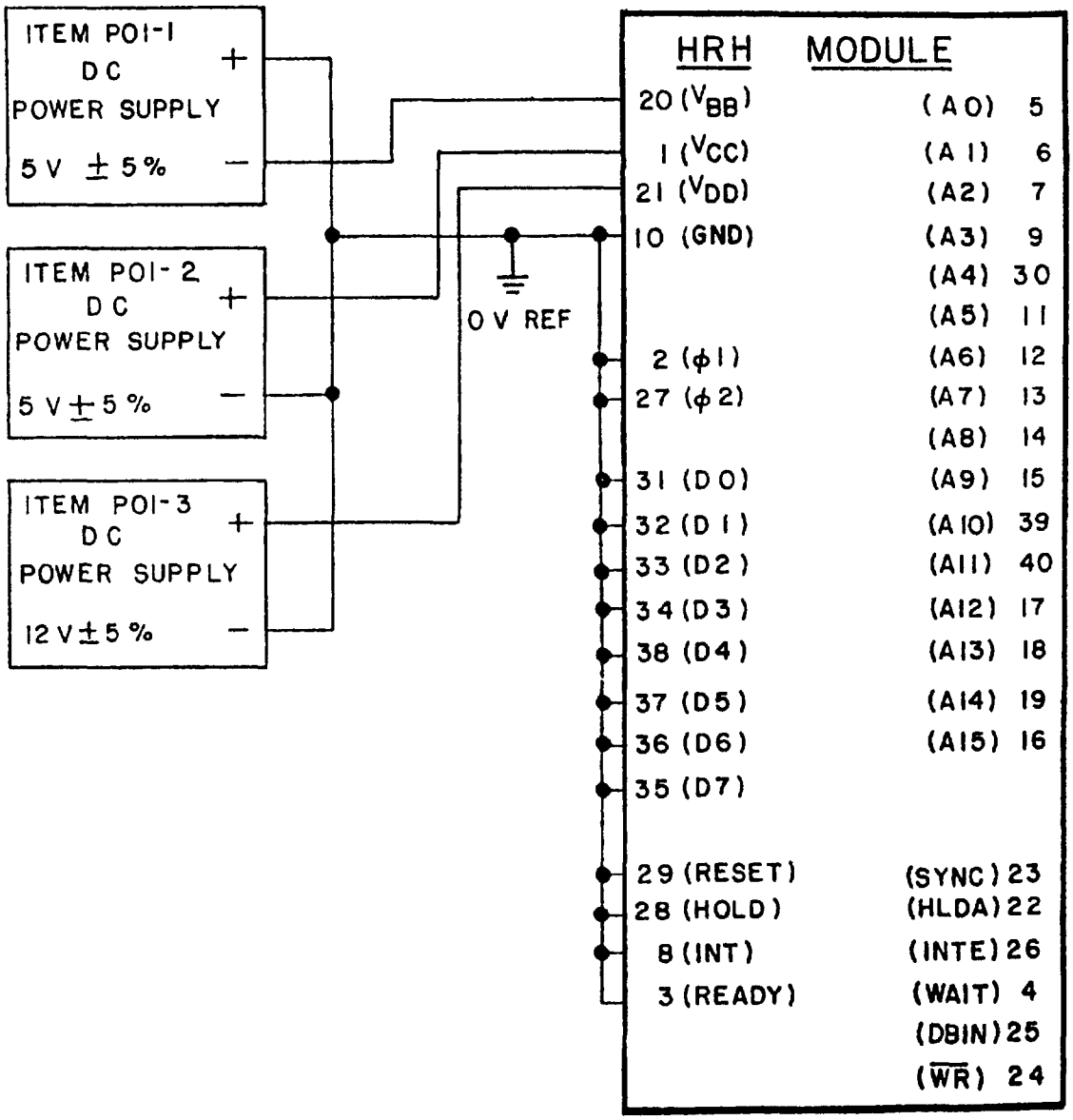


FIGURE 7. Vibration test circuit.

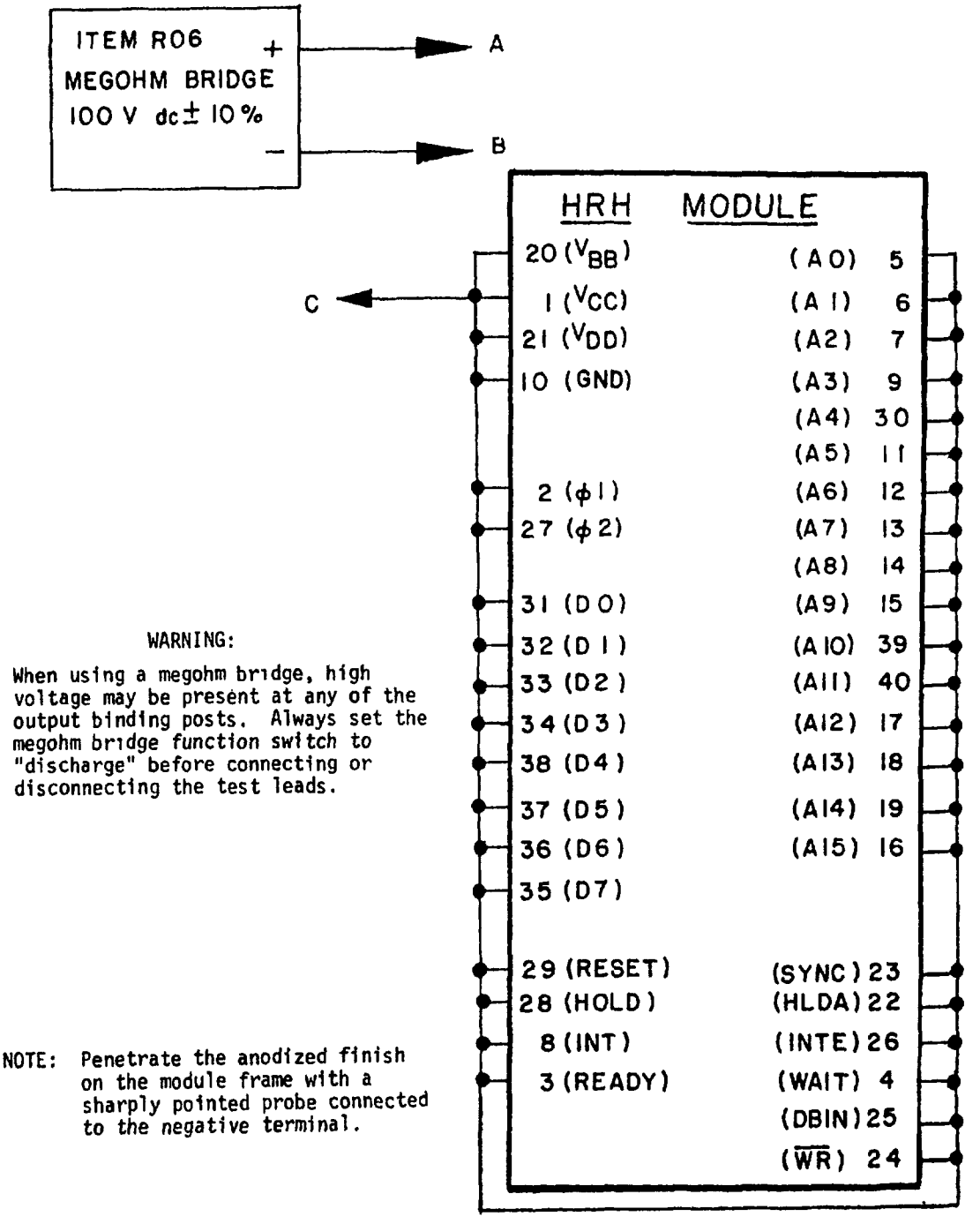
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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	C	Module frame	10	∞	MΩ	8

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

FIGURE 8. Isolation test circuit for table III.

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TABLE IV. Power supply stress check test.

Test condition	Parameter	Connect					Set in order shown	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin		
B1	V_{BB} to V_{DD} Power supply stress check	E	20	1	21	A	P01-1 to 8 V +0%, -1% P01-2 to 5 V ±1% P01-3 to 12 V +0%, -1%	9
B2	V_{CC} "	"	"	"	"	"	P01-1 to 5 V +0%, -1% P01-2 to 15 V +0%, -1% P01-3 to 12 V ±1%	"
B3	V_{DD} "	"	"	"	"	"	P01-1 to 5 V +0%, -1% P01-2 to 5 V ±1% P01-3 to 15 V +0%, -1%	"
B4	V_{BB} to input "	"	"	Open	Open	1, 21, A	P01-1 to 20 V +0%, -1%	"
B5	Reverse V_{BB} to input "	20	E	Open	Open	1, 21, B	P01-1 to 0.3 V +0%, -1%	"

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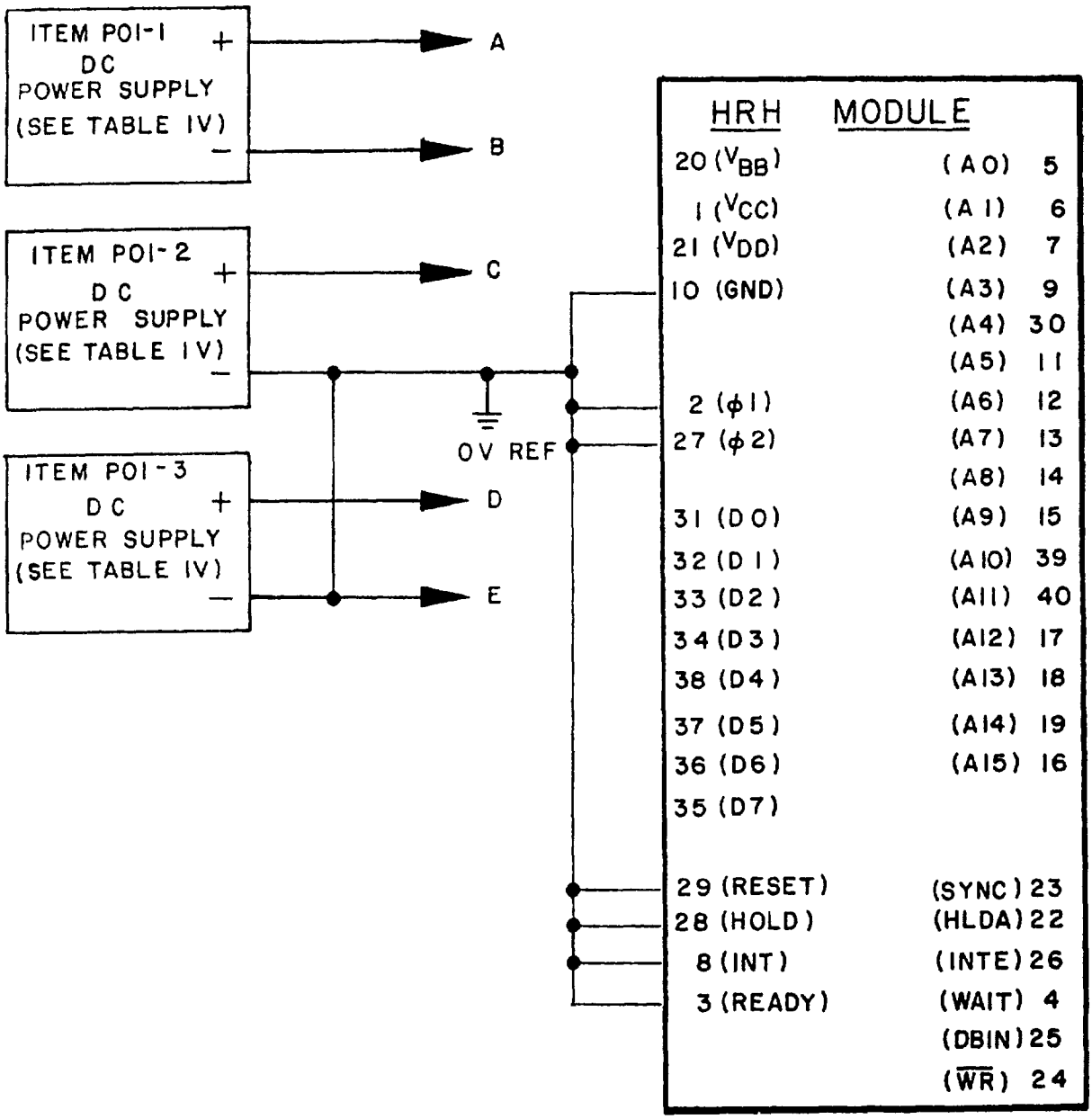


FIGURE 9. Power supply stress check test circuit for table IV.

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TABLE V. Output voltage (low-level) test.

Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
C1	A15	Output voltage, low level	16	0	450	mV	10
C2	A14	"	19	"	"	"	"
C3	A13	"	18	"	"	"	"
C4	A12	"	17	"	"	"	"
C5	A11	"	40	"	"	"	"
C6	A10	"	39	"	"	"	"
C7	A9	"	15	"	"	"	"
C8	A8	"	14	"	"	"	"
C9	A7	"	13	"	"	"	"
C10	A6	"	12	"	"	"	"
C11	A5	"	11	"	"	"	"
C12	A4	"	30	"	"	"	"
C13	A3	"	9	"	"	"	"
C14	A2	"	7	"	"	"	"
C15	A1	"	6	"	"	"	"
C16	A0	"	5	"	"	"	"
C17	D7	"	35	"	"	"	"
C18	D6	"	36	"	"	"	"
C19	D5	"	37	"	"	"	"
C20	D4	"	38	"	"	"	"
C21	D3	"	34	"	"	"	"

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TABLE V. Output voltage (low-level) test - Continued.

Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
C22	D2	Output voltage, low level	33	0	450	mV	10
C23	D1	"	32	"	"	"	"
C24	D0	"	31	"	"	"	"
C25	INTE	"	26	"	"	"	"
C26	DBIN	"	25	"	"	"	"
C27	$\overline{WR}$	"	24	"	"	"	"
C28	SYNC	"	23	"	"	"	"
C29	HLDA	"	22	"	"	"	"
C30	WAIT	"	4	"	"	"	"

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TABLE VI. Output voltage (high-level) test.

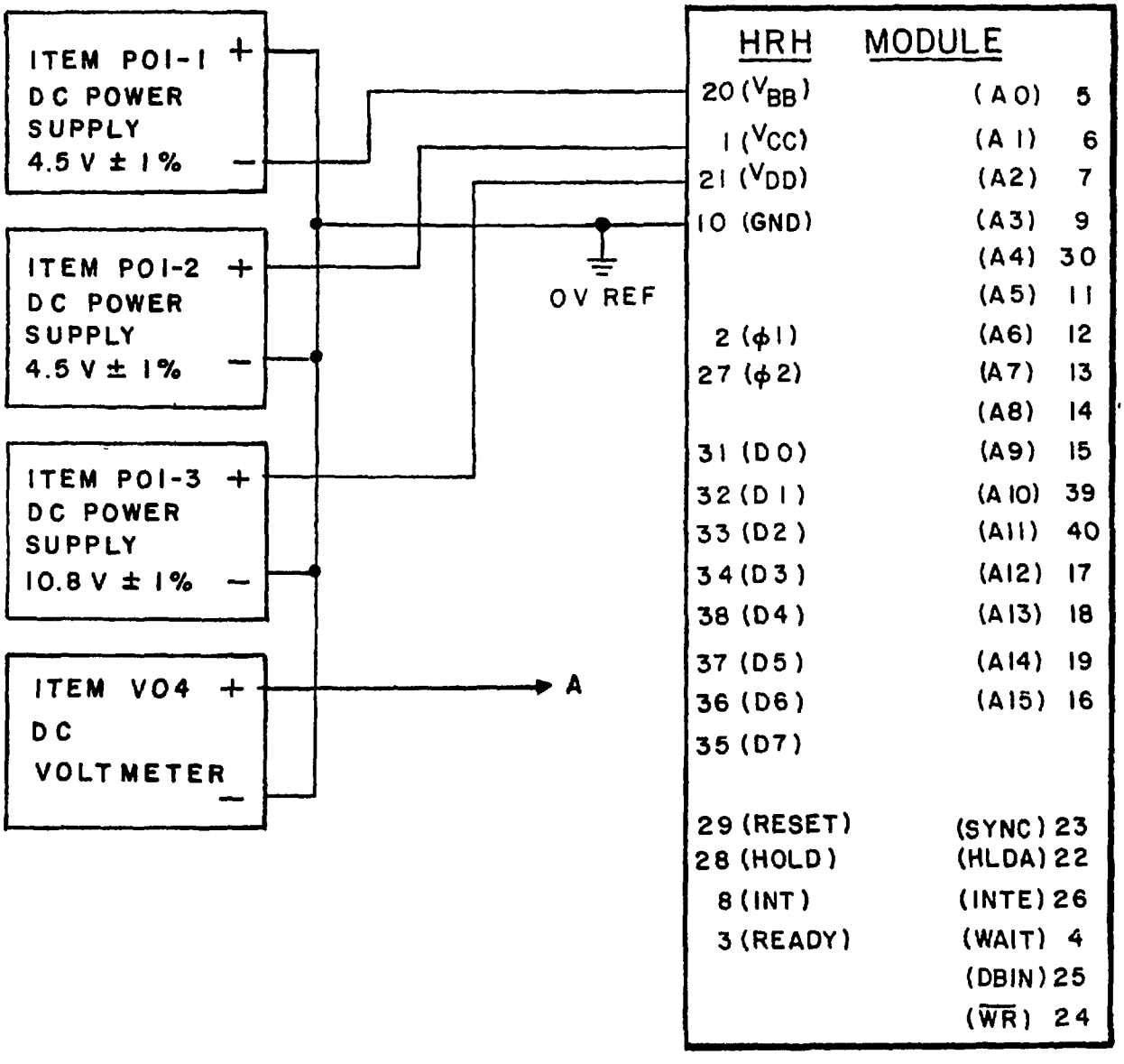
Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
D1	A15	Output voltage, high level	16	3.7	4.5	V	10
D2	A14	"	19	"	"	"	"
D3	A13	"	18	"	"	"	"
D4	A12	"	17	"	"	"	"
D5	A11	"	40	"	"	"	"
D6	A10	"	39	"	"	"	"
D7	A9	"	15	"	"	"	"
D8	A8	"	14	"	"	"	"
D9	A7	"	13	"	"	"	"
D10	A6	"	12	"	"	"	"
D11	A5	"	11	"	"	"	"
D12	A4	"	30	"	"	"	"
D13	A3	"	9	"	"	"	"
D14	A2	"	7	"	"	"	"
D15	A1	"	6	"	"	"	"
D16	A0	"	5	"	"	"	"
D17	D7	"	35	"	"	"	"
D18	D6	"	36	"	"	"	"
D19	D5	"	37	"	"	"	"
D20	D4	"	38	"	"	"	"
D21	D3	"	34	"	"	"	"

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TABLE VI. Output voltage (high-level) test - Continued.

Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
D22	D2	Output voltage, high level	33	3.7	4.5	V	10
D23	D1	"	32	"	"	"	"
D24	D0	"	31	"	"	"	"
D25	WAIT	"	4	"	"	"	"
D26	INTE	"	26	"	"	"	"
D27	DBIN	"	25	"	"	"	"
D28	$\overline{WR}$	"	24	"	"	"	"
D29	HLDA	"	22	"	"	"	"

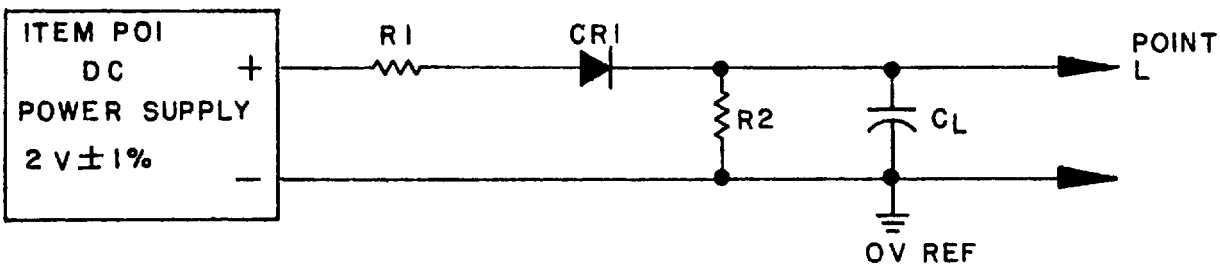
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NOTES:
1. Apply power in the following order: V_{BB}, V_{CC}, and V_{DD}.
2. Apply the preconditioning pattern in tables VII - XI using the timing conditions on figure 14.
3. Continue the waveforms and connections after the last vector is reached. Load the output as specified on figure 11 and then make the indicated connection at point A and make the measurements.
4. Since the measurements do not change the state of the microprocessor unit, measurements can continue on succeeding pins without reapplying the preconditioning pattern as long as the connections and the clock waveforms are continued within a pattern.

FIGURE 10. Output voltage (low-level and high-level) test circuit for tables V and VI.

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NOTES:
1. CR1 = 1N3600, 1N4150, or equivalent.
2. R1 = Select value for a current flow of 1.9 mA ±5% out of point L with point L held at 0.45 V.
3. R2 = 24.9 kΩ ±1%, 1/8 W.
4. C_L = 50 ±5 pF (including probe and fixture capacitance) for pins 23, 22, 26, 4, 25, 24, 31-34, 35-38, 5-7, 9, 30, 11-15, 39, 40, 17, 18, 19, and 16.
5. Lead length from load circuit to module output pin under test shall be less than 6 inches (152.4 mm).

FIGURE 11. HRH logic load circuit.

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TABLE VII. Preconditioning pattern for output voltage (low-level) on all but WAIT.

Pin	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Name	V_{CC}	Ø1	READY	WAIT	A0	A1	A2	INT	A3	GND	A5	A6	A7	A8	A9	A15	A12	A13	A14	V_{BB}
1	V_{CC}	C	A					B		GND										V_{BB}
2	"	"	"					"		"										"
3	"	"	"					"		"										"
4	"	"	"					"		"										"
5	"	"	"					"		"										"
6	"	"	"					"		"										"
7	"	"	"					"		"										"
8	"	"	"					"		"										"
9	"	"	"					"		"										"
10	"	"	"					"		"										"
11	"	"	"					"		"										"
12	"	"	"					"		"										"
13	"	"	"					"		"										"
14	"	"	"					"		"										"
15	"	"	"					"		"										"
16	"	"	"					"		"										"
17	"	"	"					"		"										"
18	"	"	"					"		"										"
19	"	"	B					"		"										"
20	"	"	B		L	L	L	"	L	"	L	L	L	L	L	L	L	L	L	"

Pin	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Name	V_{DD}	HLDA	SYNC	$\overline{WR}$	DBIN	INTE	Ø2	HOLD	RESET	A4	D0	D1	D2	D3	D7	D6	D5	D4	A10	A11
1	V_{DD}						C	B	A											
2	"						"	"	A											
3	"						"	"	A											
4	"						"	"	B											
5	"						"	"	"											
6	"						"	"	"											
7	"						"	"	"		B	A	A	A	B	B	A	A		
8	"						"	"	"											
9	"						"	"	"											
10	"						"	"	"											
11	"						"	"	"		B	B	B	B	B	B	B	B		
12	"						"	"	"											
13	"						"	"	"											
14	"						"	"	"		A	A	B	B	A	A	B	A		
15	"						"	"	"											
16																				
17																				
18											B	B	B	B	B	B	B	B		
19																				
20		L	L	L	L	L				L	L	L	L	L	L	L	L	L	L	L

A = 3 V ±1%.
B ≥ 0.8 V ±1%.
C = Clock pulse with characteristics of figure 14.
L = Low level output.
Blank = Open circuit.

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TABLE VIII. Preconditioning pattern for output voltage (low-level) on WAIT and output voltage (high-level) on HLDA.

Pin	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Name	V_{CC}	Ø1	READY	WAIT	A0	A1	A2	INT	A3	GND	A5	A6	A7	A8	A9	A15	A12	A13	A14	V_{BB}
1	V_{CC}	C	A					B		GND										V_{BB}
2	"	"	"					"		"										"
3	"	"	"					"		"										"
4	"	"	"	L				"		"										"

Pin	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Name	V_{DD}	HLDA	SYNC	$\overline{WR}$	DBIN	INTE	Ø2	HOLD	RESET	A4	D0	D1	D2	D3	D7	D6	D5	D4	A10	A11
1	V_{DD}						C	B	A											
2	"						"	B	A											
3	"						"	B	A											
4	"	H					"	A	B											

A = 3 V ±1%.
B ≥ 0.8 V ±1%.
C = Clock pulse with characteristics of figure 14.
L = Low level output.
H = High level output.
Blank = Open circuit.

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TABLE IX. Preconditioning pattern for output voltage (high-level) on all but INTE, HLDA, DBIN, and $\overline{WR}$.

Pin	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Name	V_{CC}	Ø1	READY	WAIT	A0	A1	A2	INT	A3	GND	A5	A6	A7	A8	A9	A15	A12	A13	A14	V_{BB}
1	V_{CC}	C	A					B		GND										V_{BB}
2	"	"	"					"		"										"
3	"	"	"					"		"										"
4	"	"	"					"		"										"
5	"	"	"					"		"										"
6	"	"	"					"		"										"
7	"	"	"					"		"										"
8	"	"	"					"		"										"
9	"	"	"					"		"										"
10	"	"	"					"		"										"
11	"	"	"					"		"										"
12	"	"	"					"		"										"
13	"	"	"					"		"										"
14	"	"	"					"		"										"
15	"	"	"					"		"										"
16	"	"	"					"		"										"
17	"	"	"					"		"										"
18	"	"	"					A		"										"
19	"	"	B					A		"										"
20	"	"	B	H	H	H	H	A	H	"	H	H	H	H	H	H	H	H	H	"

Pin	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Name	V_{DD}	HLDA	SYNC	$\overline{WR}$	DBIN	INTE	Ø2	HOLD	RESET	A4	D0	D1	D2	D3	D7	D6	D5	D4	A10	A11
1	V_{DD}						C	B	A											
2	"						"	"	A											
3	"						"	"	A											
4	"						"	"	B											
5	"						"	"	"											
6	"						"	"	"											
7	"						"	"	"		B	A	A	A	B	B	A	A		
8	"						"	"	"											
9	"						"	"	"											
10	"						"	"	"											
11	"						"	"	"		A	A	A	A	A	A	A	A		
12	"						"	"	"											
13	"						"	"	"											
14	"						"	"	"		A	A	B	B	A	A	B	A		
15	"						"	"	"											
16	"						"	"	"											
17	"						"	"	"											
18	"						"	"	"		A	A	A	A	A	A	A	A		
19	"						"	"	"											
20	"						"	"	"	H	H	H	H	H	H	H	H	H	H	H

A = 3 V ±1%.
B ≥ 0.8 V ±1%.
C = Clock pulse with characteristics of figure 14.
H = High level output.
Blank = Open circuit.

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TABLE X. Preconditioning pattern for output voltage (high-level) on $\overline{WR}$ and DBIN.

Pin	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Name	V_{CC}	Ø1	READY	WAIT	A0	A1	A2	INT	A3	GND	A5	A6	A7	A8	A9	A15	A12	A13	A14	V_{BB}
1	V_{CC}	C	A					B		GND										V_{BB}
2	"	"	"					"		"										"
3	"	"	"					"		"										"
4	"	"	"					"		"										"
5	"	"	"					"		"										"
6	"	"	B					"		"										"
7	"	"	B					"		"										"

Pin	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Name	V_{DD}	HLDA	SYNC	$\overline{WR}$	DBIN	INTE	Ø2	HOLD	RESET	A4	DO	D1	D2	D3	D7	D6	D5	D4	A10	A11
1	V_{DD}						C	B	A											
2	"						"	"	A											
3	"						"	"	A											
4	"						"	"	B											
5	"						"	"	"											
6	"						"	"	"											
7	"			H	H		"	"	"											

A = 3 V ±1%.
B ≥ 0.8 V ±1%.
C = Clock pulse with characteristics of figure 14.
H = High level output.
Blank = Open circuit.

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TABLE XI. Preconditioning pattern for output voltage (high-level) INTE.

Pin	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Name	V_{CC}	Ø1	READY	WAIT	A0	A1	A2	INT	A3	GND	A5	A6	A7	A8	A9	A15	A12	A13	A14	V_{BB}
1	V_{CC}	C	A					B		GND										V_{BB}
2	"	"	A					"		"										"
3	"	"	A					"		"										"
4	"	"	B					"		"										"
5	"	"	B					"		"										"
6	"	"	A					"		"										"
7	"	"	A					"		"										"

Pin	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Name	V_{DD}	HLDA	SYNC	$\overline{WR}$	DBIN	INTE	Ø2	HOLD	RESET	A4	D0	D1	D2	D3	D7	D6	D5	D4	A10	A11
1	V_{DD}						C	B	A											
2	"						"	"	A											
3	"						"	"	A											
4	"						"	"	B											
5	"						"	"	"											
6	"						"	"	"											
7	"					H	"	"	"											

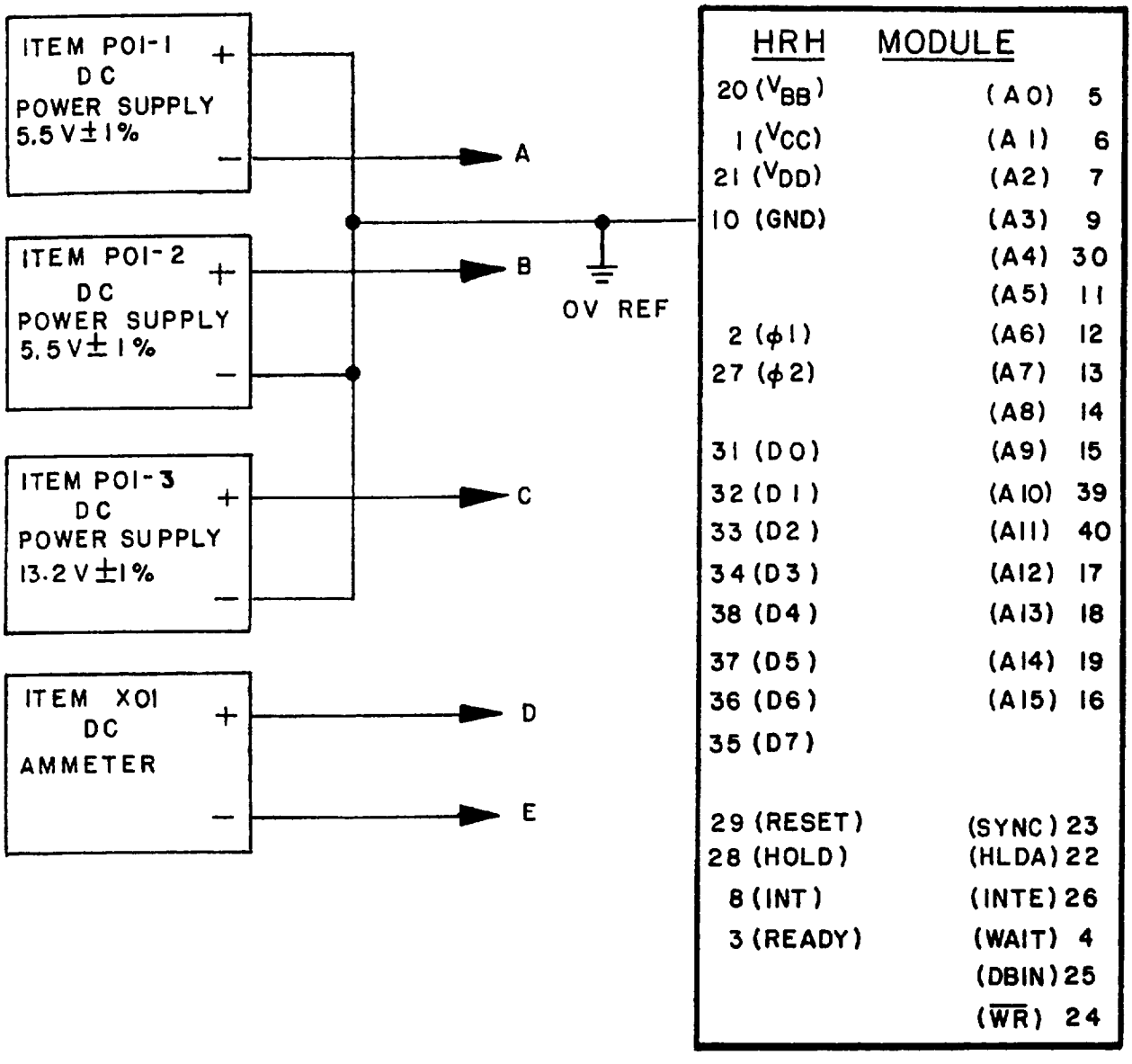
A = 3 V ±1%.
B ≥ 0.8 V ±1%.
C = Clock pulse with characteristics of figure 14.
H = High level output.
Blank = Open circuit.

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TABLE XII. Power supply current test.

Test condition	Parameter	Connect					Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Min	Max		
E1	V_{BB} Power supply current	E	1	21	20	A	0	1	mA	12
E2	V_{CC} "	20	D	21	B	1	10	100	"	"
E3	V_{DD} "	20	1	D	C	21	10	80	"	"

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CAUTION

Do not isolate the chassis of item X01 from earth ground.

FIGURE 12. Power supply current test circuit for table XII.

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TABLE XIII. Input current (high-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F1	01 Input current, high level	3,8,27,28,29	2	Open	Open	-10	10	µA	13
F2	02 "	2,3,8,28,29	27	"	"	"	"	"	"
F3	RESET "	2,3,8,27,28	29	"	"	"	"	"	"
F4	HOLD "	2,3,8,27,29	28	"	"	"	"	"	"
F5	INT "	2,3,27,28,29	8	"	"	"	"	"	"
F6	READY "	2,8,27,28,29	3	"	"	"	"	"	"
F7	DO "	32,33,34,35, 36,37,38	31	"	"	-100	100	"	"
F8	D1 "	31,33,34,35, 36,37,38	32	"	"	"	"	"	"
F9	D2 "	31,32,34,35, 36,37,38	33	"	"	"	"	"	"
F10	D3 "	31,32,33,35, 36,37,38	34	"	"	"	"	"	"
F11	D4 "	31,32,33,34, 35,36,37	38	"	"	"	"	"	"
F12	D5 "	31,32,33,34, 35,36,38	37	"	"	"	"	"	"
F13	D6 "	31,32,33,34, 35,37,38	36	"	"	"	"	"	"
F14	D7 "	31,32,33,34, 36,37,38	35	"	"	"	"	"	"

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TABLE XIV. Input current (data bus, 3.7 V level) test.

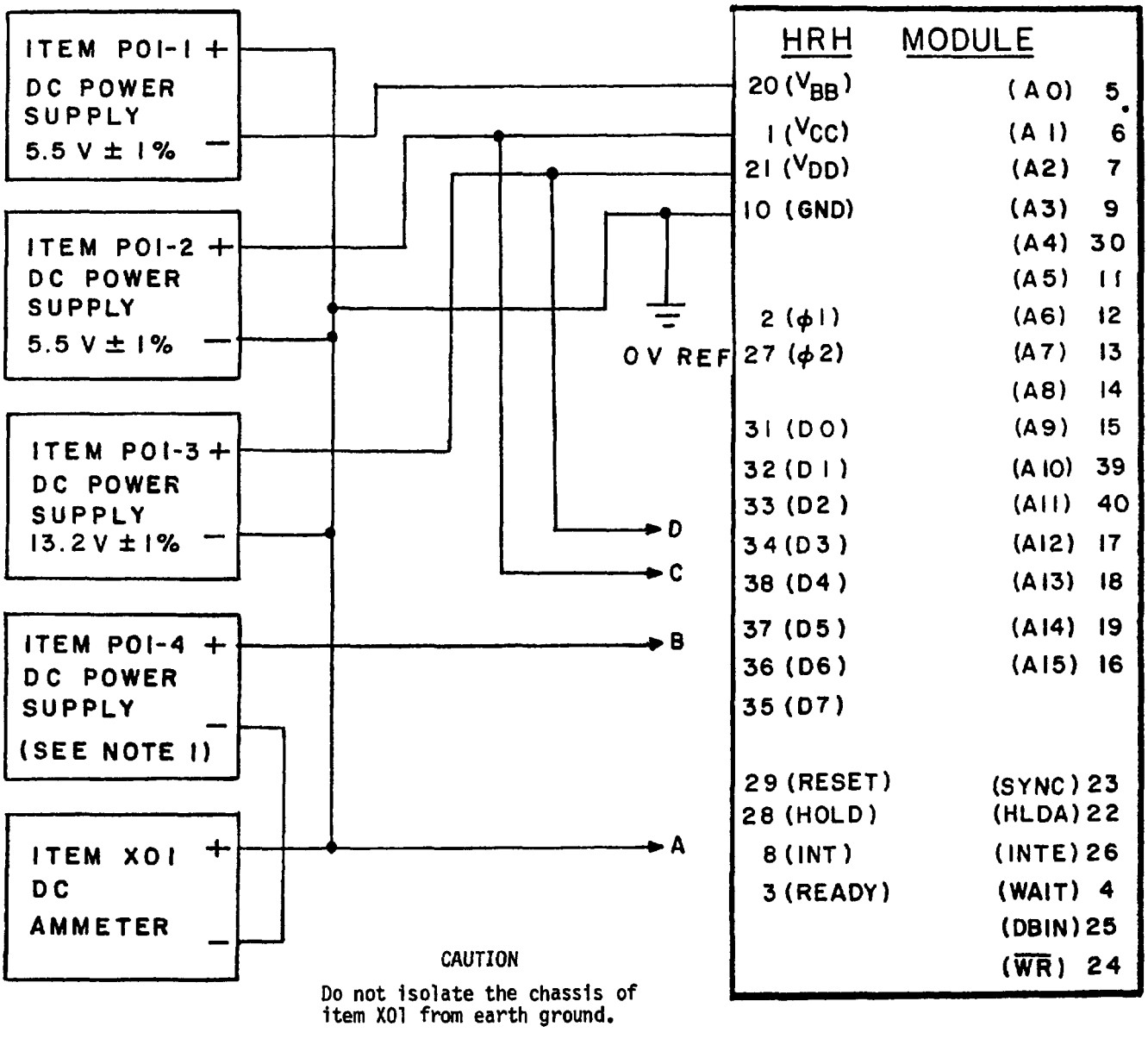
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
G1	D0 Input current, data bus, 3.7 V level	32,33,34,35, 36,37,38	31	Open	Open	-0.01	-2.00	mA	13
G2	D1 "	31,33,34,35, 36,37,38	32	"	"	"	"	"	"
G3	D2 "	31,32,34,35, 36,37,38	33	"	"	"	"	"	"
G4	D3 "	31,32,33,35, 36,37,38	34	"	"	"	"	"	"
G5	D4 "	31,32,33,34, 35,36,37	38	"	"	"	"	"	"
G6	D5 "	31,32,33,34, 35,36,38	37	"	"	"	"	"	"
G7	D6 "	31,32,33,34, 35,37,38	36	"	"	"	"	"	"
G8	D7 "	31,32,33,34, 36,37,38	35	"	"	"	"	"	"

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TABLE XV. Input current (low-level) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
H1	Ø1 Input current, low level	Open	2	3,8,28,29	27	-10	10	µA	13
H2	Ø2 "	Open	27	3,8,28,29	2	"	"	"	"
H3	RESET "	2, 27	29	3,8,28	Open	"	"	"	"
H4	HOLD "	"	28	3,8,29	"	"	"	"	"
H5	INT "	"	8	3,28,29	"	"	"	"	"
H6	READY "	"	3	8,28,29	"	"	"	"	"
H7	DO "	Open	31	32,33,34,35, 36,37,38	"	0	-100	"	"
H8	D1 "	"	32	31,33,34,35, 36,37,38	"	"	"	"	"
H9	D2 "	"	33	31,32,34,35, 36,37,38	"	"	"	"	"
H10	D3 "	"	34	31,32,33,35, 36,37,38	"	"	"	"	"
H11	D4 "	"	38	31,32,33,34, 35,36,37	"	"	"	"	"
H12	D5 "	"	37	31,32,33,34, 35,36,38	"	"	"	"	"
H13	D6 "	"	36	31,32,33,34, 35,37,38	"	"	"	"	"
H14	D7 "	"	35	31,32,33,34, 36,37,38	"	"	"	"	"

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CAUTION

Do not isolate the chassis of item XO1 from earth ground.

NOTES:
1. Item PO1-4 setting for test conditions:
   F1, F2; 13.2 V ±1%.
   F3-F14; 5.5 V ±1%.
   G1-G8; 3.7 V ±1%.
   H1-H14; 0.00 ±0.01 V.
2. For test conditions F1-F6 and H1-H6, apply power in the following order: V_{BB}, V_{CC}, V_{DD}, A, C, and D (as required). Item PO1-4 is adjusted for the required dc parametric forcing voltage, B is then connected, and the dc current is measured from point B.

* FIGURE 13. Input current (high-level, data bus, 3.7 V level, and low-level) test circuit for tables XIII, XIV, and XV.

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NOTES: Continued

3. In order to measure data bus input current, the microprocessor must be stimulated so that the data bus is in the input mode. This can be performed as follows:

   Apply power in the following order: V_{BB}, V_{CC}, and V_{DD}.

   Apply the preconditioning pattern (table XVI) and the voltages of figure 14. The timing parameters of figure 14 are used to apply the preconditioning pattern.

4. After the last vector (table XVI) is reached, the Ø1 and Ø2 signals and all other signal logic levels shall be maintained as specified for the final vector during the time that the input measurements are taken. Once this state is reached, connections shall be made points A and B and the input current parameter measured. Since the measurement of input current does not change the state of the microprocessor, successive input current measurements can continue without reapplying the preconditioning pattern as long as the conditions of the last vector are continued.

FIGURE 13. Input current (high-level, data bus, 3.7 V level, and low-level) test circuit for tables XIII, XIV, and XV - Continued.

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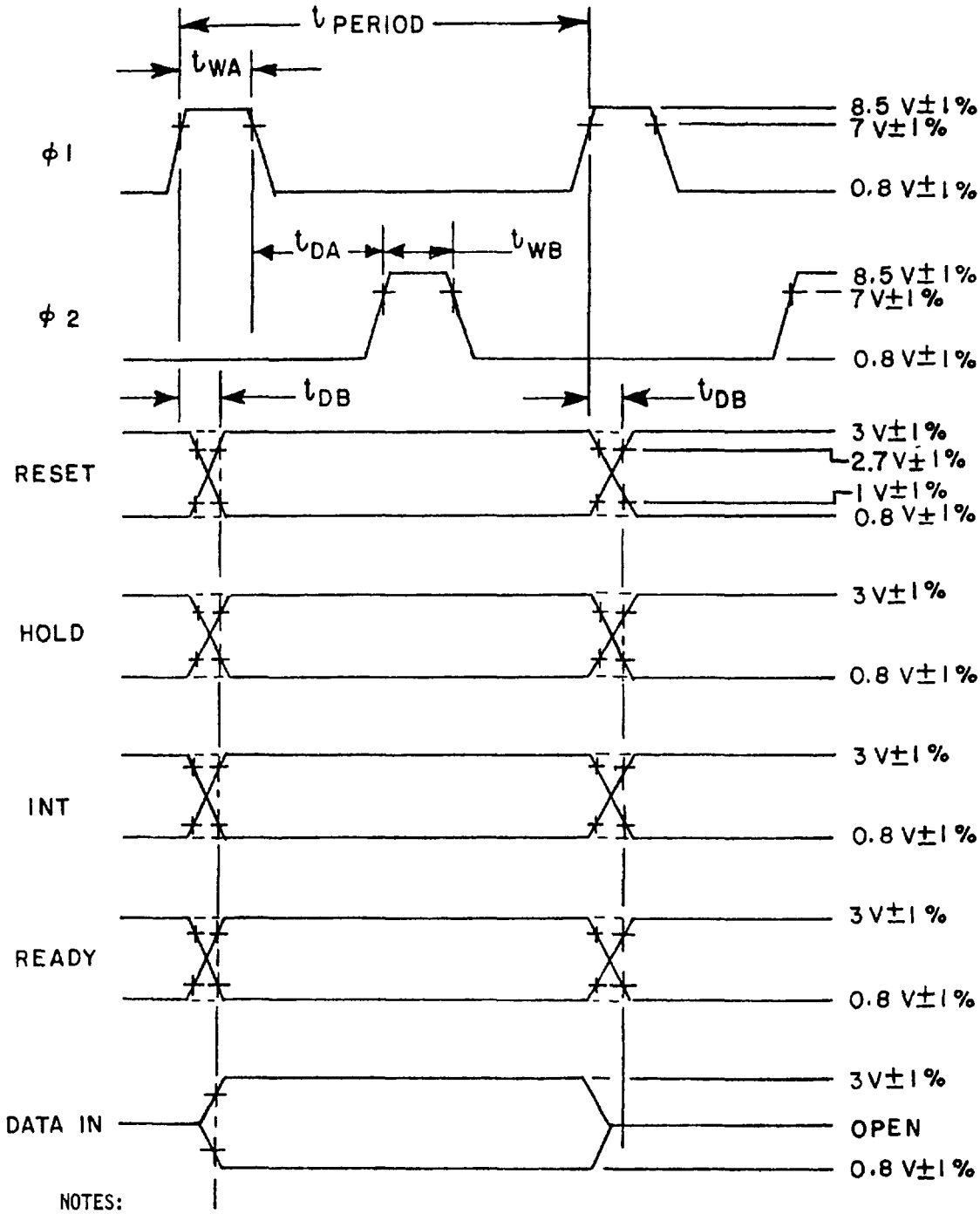
TABLE XVI. Preconditioning pattern for input current (high-level, data bus, 3.7 V level, and low-level).

Pin	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Name	V_{CC}	Ø1	READY	WAIT	A0	A1	A2	INT	A3	GND	A5	A6	A7	A8	A9	A15	A12	A13	A14	V_{BB}
1	V_{CC}	C	A					B		GND										V_{BB}
2	"	"	"					"		"										"
3	"	"	"					"		"										"
4	"	"	"					"		"										"
5	"	"	"					"		"										"
6	"	"	"					"		"										"
7	"	"	"					"		"										"
8	"	"	"					"		"										"
9	"	"	"					"		"										"
10	"	"	B					"		"										"
11	"	"	B					"		"										"

Pin	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Name	V_{DD}	HLDA	SYNC	$\overline{WR}$	DBIN	INTE	Ø2	HOLD	RESET	A4	D0	D1	D2	D3	D7	D6	D5	D4	A10	A11
1	V_{DD}						C	B	A											
2	"						"	"	A											
3	"						"	"	A											
4	"						"	"	B											
5	"						"	"	"											
6	"						"	"	"											
7	"						"	"	"		A	A	B	A	A	A	A	A		
8	"						"	"	"											
9	"						"	"	"											
10	"						"	"	"											
11	"						"	"	"		I	I	I	I	I	I	I	I		

A = 3 V ±1%.
B = 0.8 V ±1%.
C = Pulse with timing and voltage characteristics as detailed on figure 14.
Blank = Open circuit.
I = Data bus in input mode.

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NOTES:
1. Pulse parameters:
   t_{PERIOD} = 2 µs ±20 ns.
   t_{WA} = 150 ±1 ns.
   t_{WB} = 220 ±1 ns.
   t_{DA} = t_{DB} = 0 +2, -0 ns.
2. RESET, HOLD, INT, READY, and DATA BUS are shown changing level (when specified by table XIX) coincident with the leading edge of Ø1. These times, along with t_{WA}, t_{WB}, and t_{DA}, will provide the desired setup conditions.

FIGURE 14. Output voltage, input current, and three-state output leakage current preconditioning vector waveforms for tables V, VI, XIII, XIV, XV, XVII, and XVIII.

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TABLE XVII. Three-state output leakage current (low-level) test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
J1	D0	Three-state output leakage current, low level	5,6,7,9,11,12,13,14,15, 16,17,18,19,30,32,33, 34,35,36,37,38,39,40	31	0	-100	µA	15
J2	D1	"	5,6,7,9,11,12,13,14,15, 16,17,18,19,30,31,33, 34,35,36,37,38,39,40	32	"	"	"	"
J3	D2	"	5,6,7,9,11,12,13,14,15, 16,17,18,19,30,31,32, 34,35,36,37,38,39,40	33	"	"	"	"
J4	D3	"	5,6,7,9,11,12,13,14,15, 16,17,18,19,30,31,32, 33,35,36,37,38,39,40	34	"	"	"	"
J5	D4	"	5,6,7,9,11,12,13,14,15, 16,17,18,19,30,31,32, 33,34,35,36,37,39,40	38	"	"	"	"
J6	D5	"	5,6,7,9,11,12,13,14,15, 16,17,18,19,30,31,32, 33,34,35,36,38,39,40	37	"	"	"	"
J7	D6	"	5,6,7,9,11,12,13,14,15, 16,17,18,19,30,31,32, 33,34,35,37,38,39,40	36	"	"	"	"
J8	D7	"	5,6,7,9,11,12,13,14,15, 16,17,18,19,30,31,32, 33,34,36,37,38,39,40	35	"	"	"	"
J9	A0	"	6,7,9,11,12,13,14,15, 16,17,18,19,30,31,32, 33,34,35,36,37,38,39,40	5	"	"	"	"

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TABLE XVII. Three-state output leakage current (low-level) test - Continued.

Test condition		Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
J10	A1	Three-state output leakage current, low level	5,7,9,11,12,13,14,15,16, 17,18,19,30,31,32,33,34, 35,36,37,38,39,40	6	0	-100	ΩA	15
J11	A2	"	5,6,9,11,12,13,14,15,16, 17,18,19,30,31,32,33,34, 35,36,37,38,39,40	7	"	"	"	"
J12	A3	"	5,6,7,11,12,13,14,15,16, 17,18,19,30,31,32,33,34, 35,36,37,38,39,40	9	"	"	"	"
J13	A4	"	5,6,7,9,11,12,13,14,15, 16,17,18,19,31,32,33,34, 35,36,37,38,39,40	30	"	"	"	"
J14	A5	"	5,6,7,9,12,13,14,15,16, 17,18,19,30,31,32,33,34, 35,36,37,38,39,40	11	"	"	"	"
J15	A6	"	5,6,7,9,11,13,14,15,16, 17,18,19,30,31,32,33,34, 35,36,37,38,39,40	12	"	"	"	"
J16	A7	"	5,6,7,9,11,12,14,15,16, 17,18,19,30,31,32,33,34, 35,36,37,38,39,40	13	"	"	"	"
J17	A8	"	5,6,7,9,11,12,13,15,16, 17,18,19,30,31,32,33,34, 35,36,37,38,39,40	14	"	"	"	"
J18	A9	"	5,6,7,9,11,12,13,14,16, 17,18,19,30,31,32,33,34, 35,36,37,38,39,40	15	"	"	"	"

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TABLE XVII. Three-state output leakage current (low-level) test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
J19	A10	Three-state output leakage current, low level	5,6,7,9,11,12,13,14,15, 16,17,18,19,30,31,32,33, 34,35,36,37,38,40	39	0	-100	µA	15
J20	A11	"	5,6,7,9,11,12,13,14,15, 16,17,18,19,30,31,32,33, 34,35,36,37,38,39	40	"	"	"	"
J21	A12	"	5,6,7,9,11,12,13,14,15, 16,18,19,30,31,32,33,34, 35,36,37,38,39,40	17	"	"	"	"
J22	A13	"	5,6,7,9,11,12,13,14,15, 16,17,19,30,31,32,33,34, 35,36,37,38,39,40	18	"	"	"	"
J23	A14	"	5,6,7,9,11,12,13,14,15, 16,17,18,30,31,32,33,34, 35,36,37,38,39,40	19	"	"	"	"
J24	A15	"	5,6,7,9,11,12,13,14,15, 17,18,19,30,31,32,33,34, 35,36,37,38,39,40	16	"	"	"	"

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TABLE XVIII. Three-state output leakage current (high-level) test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
K1	D0	Three-state output leakage current, high level	5,6,7,9,11,12,13,14,15, 16,17,18,19,30,32,33, 34,35,36,37,38,39,40	31	0	10	µA	15
K2	D1	"	5,6,7,9,11,12,13,14,15, 16,17,18,19,30,31,33, 34,35,36,37,38,39,40	32	"	"	"	"
K3	D2	"	5,6,7,9,11,12,13,14,15, 16,17,18,19,30,31,32, 34,35,36,37,38,39,40	33	"	"	"	"
K4	D3	"	5,6,7,9,11,12,13,14,15, 16,17,18,19,30,31,32, 33,35,36,37,38,39,40	34	"	"	"	"
K5	D4	"	5,6,7,9,11,12,13,14,15, 16,17,18,19,30,31,32, 33,34,35,36,37,39,40	38	"	"	"	"
K6	D5	"	5,6,7,9,11,12,13,14,15, 16,17,18,19,30,31,32, 33,34,35,36,38,39,40	37	"	"	"	"
K7	D6	"	5,6,7,9,11,12,13,14,15, 16,17,18,19,30,31,32, 33,34,35,37,38,39,40	36	"	"	"	"
K8	D7	"	5,6,7,9,11,12,13,14,15, 16,17,18,19,30,31,32, 33,34,36,37,38,39,40	35	"	"	"	"
K9	A0	"	6,7,9,11,12,13,14,15, 16,17,18,19,30,31,32, 33,34,35,36,37,38,39,40	5	"	"	"	"

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TABLE XVIII. Three-state output leakage current (high-level) test - Continued.

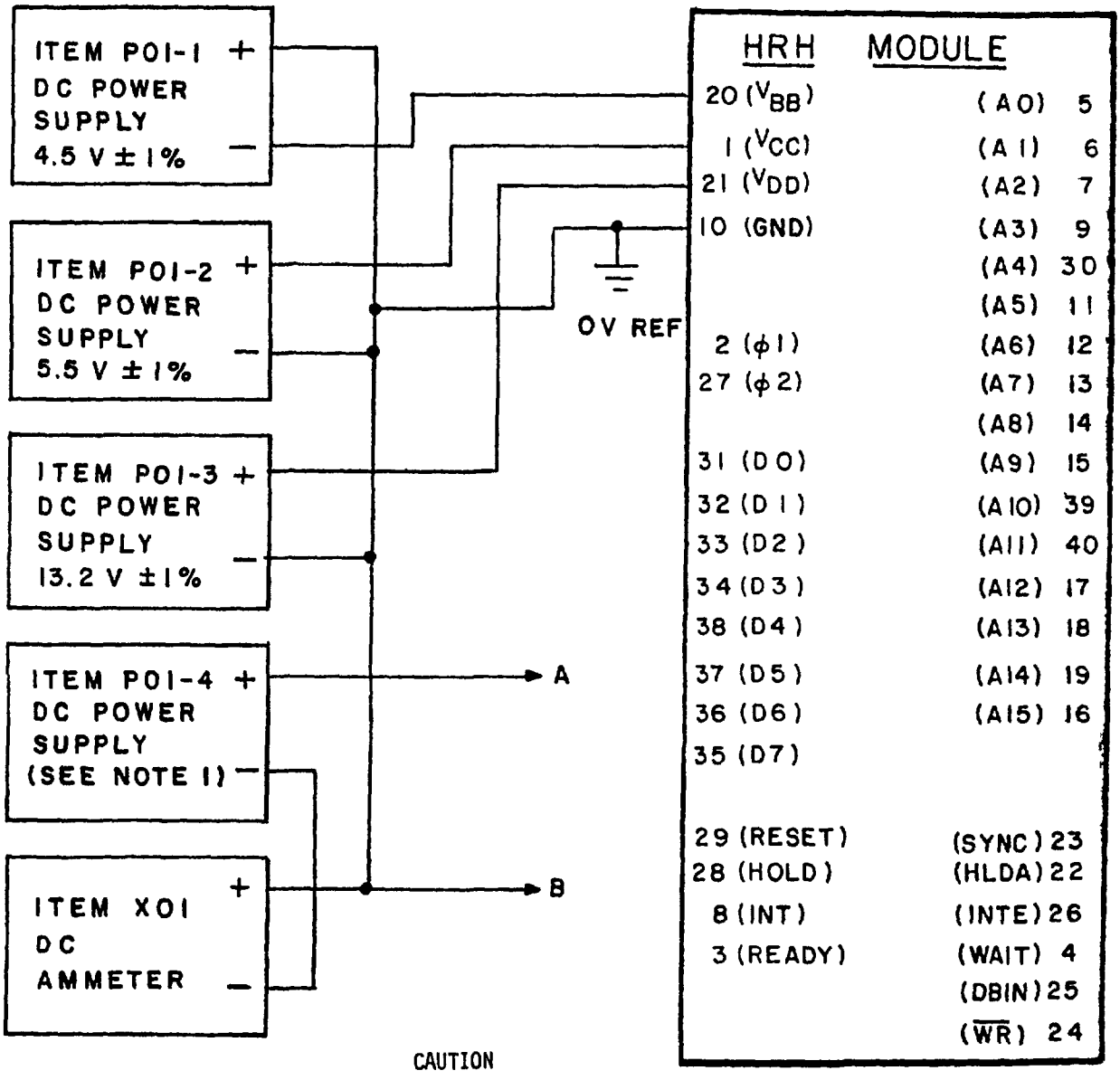
Test condition		Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
K10	A1	Three-state output leakage current, high level	5,7,9,11,12,13,14,15,16, 17,18,19,30,31,32,33,34, 35,36,37,38,39,40	6	0	10	µA	15
K11	A2	"	5,6,9,11,12,13,14,15,16, 17,18,19,30,31,32,33,34, 35,36,37,38,39,40	7	"	"	"	"
K12	A3	"	5,6,7,11,12,13,14,15,16, 17,18,19,30,31,32,33,34, 35,36,37,38,39,40	9	"	"	"	"
K13	A4	"	5,6,7,9,11,12,13,14,15, 16,17,18,19,31,32,33, 34,35,36,37,38,39,40	30	"	"	"	"
K14	A5	"	5,6,7,9,12,13,14,15,16, 17,18,19,30,31,32,33,34, 35,36,37,38,39,40	11	"	"	"	"
K15	A6	"	5,6,7,9,11,13,14,15,16, 17,18,19,30,31,32,33,34, 35,36,37,38,39,40	12	"	"	"	"
K16	A7	"	5,6,7,9,11,12,14,15,16, 17,18,19,30,31,32,33,34, 35,36,37,38,39,40	13	"	"	"	"
K17	A8	"	5,6,7,9,11,12,13,15,16, 17,18,19,30,31,32,33,34, 35,36,37,38,39,40	14	"	"	"	"
K18	A9	"	5,6,7,9,11,12,13,14,16, 17,18,19,30,31,32,33,34, 35,36,37,38,39,40	15	"	"	"	"

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TABLE XVIII. Three-state output leakage current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K19	A10 Three-state output leakage current, high level	5,6,7,9,11,12,13,14,15, 16,17,18,19,30,31,32,33, 34,35,36,37,38,40	39	0	10	µA	15
K20	A11 "	5,6,7,9,11,12,13,14,15, 16,17,18,19,30,31,32,33, 34,35,36,37,38,39	40	"	"	"	"
K21	A12 "	5,6,7,9,11,12,13,14,15, 16,18,19,30,31,32,33,34, 35,36,37,38,39,40	17	"	"	"	"
K22	A13 "	5,6,7,9,11,12,13,14,15, 16,17,19,30,31,32,33,34, 35,36,37,38,39,40	18	"	"	"	"
K23	A14 "	5,6,7,9,11,12,13,14,15, 16,17,18,30,31,32,33,34, 35,36,37,38,39,40	19	"	"	"	"
K24	A15 "	5,6,7,9,11,12,13,14,15, 17,18,19,30,31,32,33,34, 35,36,37,38,39,40	16	"	"	"	"

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CAUTION

Do not isolate the chassis of item XOI from earth ground.

NOTES:
1. Item POI-4 setting for test conditions:
   J1-J24; 0.45 V ±1%.
   K1-K24; 5.5 V ±1%.
2. Apply power in the following order: V_{BB}, V_{CC}, and V_{DD}.
3. Apply the preconditioning pattern in table XIX using the timing conditions of figure 14.
4. Continue the waveforms and connections after the last vector is reached. Make the inidicated connections at points A and B and make the measurements.
5. Since the measurements do not change the state of the microprocessor, measurements can continue on succeeding pins without reapplying the preconditioning pattern as long as the connections and the clock waveforms are continued.

FIGURE 15. Three-state output leakage current (low-level and high-level) test circuit for tables XVII and XVIII.

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TABLE XIX. Preconditioning pattern for three-state output leakage current (low-level and high-level) for tables XVII and XVIII.

Pin	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Name	V_{CC}	Ø1	READY	WAIT	A0	A1	A2	INT	A3	GND	A5	A6	A7	A8	A9	A15	A12	A13	A14	V_{BB}
1	V_{CC}	C	A					B		GND										V_{BB}
2	"	"	"					"		"										"
3	"	"	"					"		"										"
4	"	"	"					"		"										"
5	"	"	"					"		"										"
6	"	"	"					"		"										"
7	"	"	"					"		"										"
8	"	"	"		Z	Z	Z	"	Z	"	Z	Z	Z	Z	Z	Z	Z	Z	Z	"

Pin	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
Name	V_{DD}	HLDA	SYNC	$\overline{WR}$	DBIN	INTE	Ø2	HOLD	RESET	A4	D0	D1	D2	D3	D7	D6	D5	D4	A10	A11
1	V_{DD}						C	B	A											
2	"						"	"	A											
3	"						"	"	A											
4	"						"	"	B											
5	"						"	A	"											
6	"						"	"	"											
7	"						"	"	"		B	B	B	B	B	B	B	B		
8	"						"	"	"	Z	Z	Z	Z	Z	Z	Z	Z	Z	Z	Z

A = 3 V ±1%.
B = 0.8 V ±1%.
C = Pulse with timing and voltage characteristics as detailed on figure 16.
Blank = Open circuit.
Z = Output pins in high-impedance state.

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TABLE XX. Dynamic functional test. 1/, 2/

Test condition	Test description	Input condition table	Waveform figures
L1	Dynamic functional 3/ a	XXI	16,17,18,19,20,21, 22,23,24,25,26
L2	Dynamic functional 3/ b	XXII	"
L3	Dynamic functional 3/ c	XXIII	"
L4	Dynamic functional 3/ d	XXIV	"

1/ Load the outputs as specified on figure 11.
2/ Connect power supplies in the following order: V_{BB}, V_{CC}, and V_{DD}.
3/ Test conditions L1 through L4:

a. Test condition L1 applies to the functional test vectors (see 6.5) and checks for logic outputs as detailed in note 4/ c and for the following operating conditions and parameters:

Maximum frequency of operation.
Minimum O1 to O2 clock pulse delay.
Most negative power supply tolerances.
Minimum data bus transfer time (an internal propagation delay time check).
Worst case input setup and hold times and input threshold voltages.
Loaded output voltage levels.

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b. Test condition L2 applies to the functional test vectors (see 6.5) and checks for logic outputs as detailed in 4/ c and for the following operating conditions and parameters:

Minimum operating frequency.
Minimum O1 to O2 clock pulse delay.
Most negative power supply tolerances.
Minimum data bus precharge time coupled with maximum data transfer time to the static registers.
Worst case input setup and hold times and input threshold voltages.
Loaded output voltage levels.

c. Test condition L3 applies to the functional test vectors (see 6.5) and checks for logic outputs as detailed in 4/ c and for the following operating conditions and parameters:

Minimum operating frequency.
Minimum O1 to O2 clock pulse delay.
Maximum O2 pulse width.
Worst case write conditions for dynamic accumulaltor.
Most negative power supply tolerances.
Worst case input setup and hold times and input threshold voltages.
Loaded output voltage levels.

d. Test condition L4 applies to the functional test vectors (see 6.5) and checks for logic outputs as detailed in 4/ c and for the following operating conditions and parameters:

Maximum operating frequency.
Most positive power supply tolerances (worst case high-level threshold).
Minimum O1 pulse width.
Data setup to trailing edge of O1.
Worst case input setup and hold times and input threshold voltages.
Loaded output voltage levels.

4/ The functional test vectors (see 6.5) shall be applied in the specified order with the following constraints:

a. Cycle definition. A cycle is the period beginning with a O1 leading edge and continuing until the next O1 leading edge. The input logic levels and expected output levels for a cycle (full or part of the cycle as defined below) are defined by one line (vector) (see 6.5).

b. Inputs. The basic cycle period is defined by O1. The O2 timing is referenced to O1. The other input timing conditions are in turn referenced to O1 or O2. Tables XXI through XXIV specify the input voltage and timing conditions and also refer to the waveform figures which specify the waveform edges and voltage levels to which the timing parameters are referenced. Inputs are specified (see 6.5) by either A = V_{IH}, B = V_{IL} or C = clock pulse. Clock inputs shall be driven by a positive pulse from V_{IL1} to V_{IH1} to V_{IL1} with the waveshape and timing as specified by the input timing tables and figure 16. The data bus is in the input mode whenever DBIN = 0 (see 6.5). The data bus inputs shall be driven by a return to complement waveform during these cycles as shown on figure 17. That is, the data bus inputs shall be at A (B) level from t_{SLH2} or t_{SLH3} (t_{SHL2} or t_{SHL3}) to t_{HHL2} (t_{HLH2}) and driven to B (A) otherwise during the cycle. The intent of the HOLD (figures 20 and 21), INT (figures 22 and 23), and READY (figures 17, 18, and 19) are most easily met by also driving these signals with a return to complement waveform. RESET shall be a simple NRZ (nonreturn to zero) signal from t_{SLH9} to t_{HHL3} and t_{HHL3} to t_{SLH9}.

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c. Output levels and valid cycles. Outputs shall be monitored for logic level as follows:

H $\geq$ 3 V.
L $\leq$ 0.8 V.
Z $\leq$ 1.5 V and $\geq$1 V (output pin is in high-impedance mode and the level is determined by load circuit per figure 11).
X = don't care. No measurement is required, however, the pin(s) must not be driven by the test equipment during this cycle.

d. The vector set is organized in groups where each group is less than or equal to 1,024 vectors. This enables the use of this vector set on testers with a local memory greater than or equal to 1,024 locations. Each group of vectors must be applied in the order shown without pauses. The last vector of each group is as follows:

Group number	Last vector number
1	1,024
2	1,837
3	2,861
4	3,883
5	4,905
6	5,927
7	6,949
8	7,484
9	8,508
10	9,530
11	10,552
12	10,768
13	11,761

The groups must be run consecutively in the order shown. Between application of any two groups, the user has the option of:

(1) No pause between the groups or vectors.

(2) Repeated application of the last vector of the group at test speed until the tester is ready with the next group. The outputs need not be monitored after the first application of the last vector in each group until the next group is ready.

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TABLE XXI. Input conditions for test condition L1 of table XX.

Parameter	Pins	Input condition 1/	Unit	Figure
V_{BB}	20	-5.5 ±1%	V	---
V_{CC}	1	4.5 ±1%	"	---
V_{DD}	21	10.8 ±1%	"	---
V_{IH1}	All logic inputs except clocks	3.0 ±1%	"	---
V_{IH2}	2, 27	8.5 ±1%	"	---
V_{IL1}	All logic inputs including clocks	0.8 ±1%	"	---
t_{PERIOD}	2	480	ns	16
t_{01}	2	80	"	16
t_{02}	27	220	"	16
t_{SLH1}	3	120	"	19
t_{SHL1}	3	120	"	17, 18
t_{SLH2}	31, 32, 33, 34, 38, 37, 36, 35	130	"	17, 19
t_{SHL2}	31, 32, 33, 34, 38, 37, 36, 35	130	"	17, 19
t_{SLH3}	31, 32, 33, 34, 38, 37, 36, 35	130	"	17, 19
t_{SHL3}	31, 32, 33, 34, 38, 37, 36, 35	130	"	17, 19
t_{SLH4}	8	120	"	22, 23
t_{SHL4}	8	120	"	22, 23
t_{SLH5}	28	140	"	21
t_{SHL5}	28	140	"	20
t_{SLH6}	27	0	"	16
t_{SLH7}	27	80	"	16
t_{SHL8}	27	180	"	16
t_{SLH9}	29	75	"	26
t_{HLH1}	3, 8, 28	0	"	17,18,20, 22,23
t_{HHL1}	3, 8, 28	0	"	19,21,22, 23
t_{HLH2}	31, 32, 33, 34, 38, 37, 36, 35	50	"	17,19
t_{HHL2}	31, 32, 33, 34, 38, 37, 36, 35	50	"	17,19
t_{HHL3}	29	75	"	26

1/ All input timing parameters are ±5 ns.

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TABLE XXII. Input conditions for test condition L2 of table XX.

Parameter	Pins	Input condition 1/	Unit	Figure
V_{BB}	20	-5.5 ±1%	V	---
V_{CC}	1	4.5 ±1%	"	---
V_{DD}	21	10.8 ±1%	"	---
V_{IH1}	All logic inputs except clocks	3 ±1%	"	---
V_{IH2}	2, 27	8.5 ±1%	"	---
V_{IL1}	All logic inputs including clocks	0.8 ±1%	"	---
t_{PERIOD}	2	2000	ns	16
t_{O1}	2	80	"	16
t_{O2}	27	220	"	16
t_{SLH1}	3	120	"	19
t_{SHL1}	3	120	"	17,18
t_{SLH2}	31, 32, 33, 34, 38, 37, 36, 35	130	"	17,19
t_{SHL2}	31, 32, 33, 34, 38, 37, 36, 35	130	"	17,19
t_{SLH3}	31, 32, 33, 34, 38, 37, 36, 35	130	"	17,19
t_{SHL3}	31, 32, 33, 34, 38, 37, 36, 35	130	"	17,19
t_{SLH4}	8	120	"	22,23
t_{SHL4}	8	120	"	22,23
t_{SLH5}	28	140	"	21
t_{SHL5}	28	140	"	20
t_{SLH6}	27	0	"	16
t_{SLH7}	27	80	"	16
t_{SHL8}	27	1700	"	16
t_{SLH9}	29	75	"	26
t_{HLH1}	3, 8, 28	0	"	17,18,20, 22,23
t_{HHL1}	3, 8, 28	0	"	19,21,22, 23
t_{HLH2}	31, 32, 33, 34, 38, 37, 36, 35	50	"	17,19
t_{HHL2}	31, 32, 33, 34, 38, 37, 36, 35	50	"	17,19
t_{HHL3}	29	75	"	26

1/ All input timing parameters are ±5 ns.

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TABLE XXIII. Input conditions for test condition L3 of table XX.

Parameter	Pins	Input 1/ condition	Unit	Figure
V_{BB}	20	-5.5 ±1%	V	---
V_{CC}	1	4.5 ±1%	"	---
V_{DD}	21	10.8 ±1%	"	---
V_{IH1}	All logic inputs except clocks	3 ±1%	"	---
V_{IH2}	2, 27	8.5 ±1%	"	---
V_{IL1}	All logic inputs including clocks	0.8 ±1%	"	---
t_{PERIOD}	2	2000	ns	16
t_{O1}	2	80	"	16
t_{O2}	27	1840	"	16
t_{SLH1}	3	120	"	19
t_{SHL1}	3	120	"	17,18
t_{SLH2}	31, 32, 33, 34, 38, 37, 36, 35	130	"	17,19
t_{SHL2}	31, 32, 33, 34, 38, 37, 36, 35	130	"	17,19
t_{SLH3}	31, 32, 33, 34, 38, 37, 36, 35	130	"	17,19
t_{SHL3}	31, 32, 33, 34, 38, 37, 36, 35	130	"	17,19
t_{SLH4}	8	120	"	22,23
t_{SHL4}	8	120	"	22,23
t_{SLH5}	28	140	"	21
t_{SHL5}	28	140	"	20
t_{SLH6}	27	0	"	16
t_{SLH7}	27	80	"	16
t_{SHL8}	27	80	"	16
t_{SLH9}	29	75	"	26
t_{HLH1}	3, 8, 28	0	"	17,18,20, 22,23
t_{HHL1}	3, 8, 28	0	"	19,21,22, 23
t_{HLH2}	31, 32, 33, 34, 38, 37, 36, 35	50	"	17,19
t_{HHL2}	31, 32, 33, 34, 38, 37, 36, 35	50	"	17,19
t_{HHL3}	29	75	"	26

1/ All input timing parameters are ±5 ns.

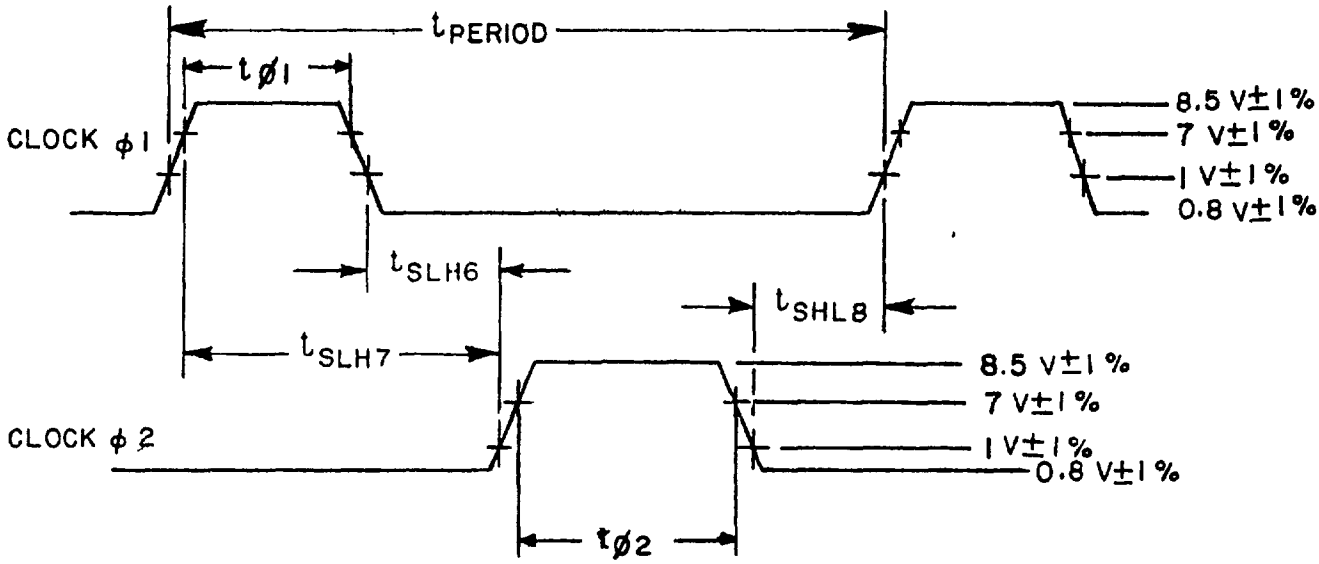
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TABLE XXIV. Input conditions for test condition L4 of table XX.

Parameter	Pins	Input 1/ condition	Unit	Figure
V_{BB}	20	-4.5 ±1%	V	---
V_{CC}	1	5.5 ±1%	"	---
V_{DD}	21	13.2 ±1%	"	---
V_{IH1}	All logic inputs except clocks	3 ±1%	"	---
V_{IH2}	2, 27	8.5 ±1%	"	---
V_{IL1}	All logic inputs including clocks	0.8 ±1%	"	---
t_{PERIOD}	2	500	ns	16
t_{O1}	2	60	"	16
t_{O2}	27	220	"	16
t_{SLH1}	3	120	"	19
t_{SHL1}	3	120	"	17,18
t_{SLH2}	31, 32, 33, 34, 38, 37, 36, 35	30	"	17,19
t_{SHL2}	31, 32, 33, 34, 38, 37, 36, 35	30	"	17,19
t_{SLH3}	31, 32, 33, 34, 38, 37, 36, 35	150	"	17,19
t_{SHL3}	31, 32, 33, 34, 38, 37, 36, 35	150	"	17,19
t_{SLH4}	8	120	"	22,23
t_{SHL4}	8	120	"	22,23
t_{SLH5}	28	140	"	21
t_{SHL5}	28	140	"	20
t_{SLH6}	27	120	"	16
t_{SLH7}	27	180	"	16
t_{SHL8}	27	100	"	16
t_{SLH9}	29	75	"	26
t_{HLH1}	3, 8, 28	0	"	17,18,20, 22,23
t_{HHL1}	3, 8, 28	0	"	19,21,22, 23
t_{HLH2}	31, 32, 33, 34, 38, 37, 36, 35	50	"	17,19
t_{HHL2}	31, 32, 33, 34, 38, 37, 36, 35	50	"	17,19
t_{HHL3}	29	75	"	26

1/ All input timing parameters are ±5 ns.

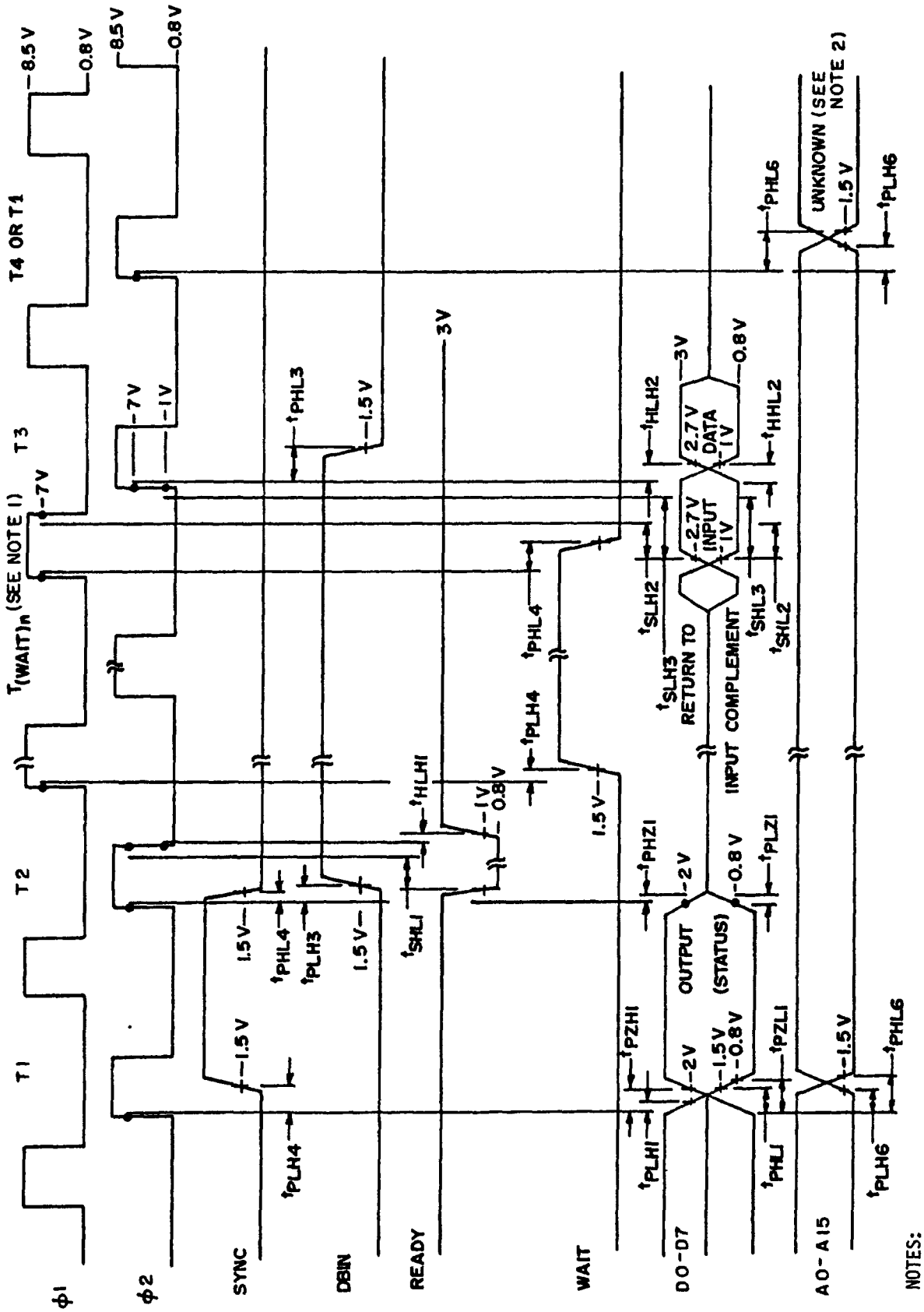
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NOTE: Clock $t_{TLH} = t_{THL} = 25 \pm 25$ ns.

* FIGURE 16. Clock waveforms.

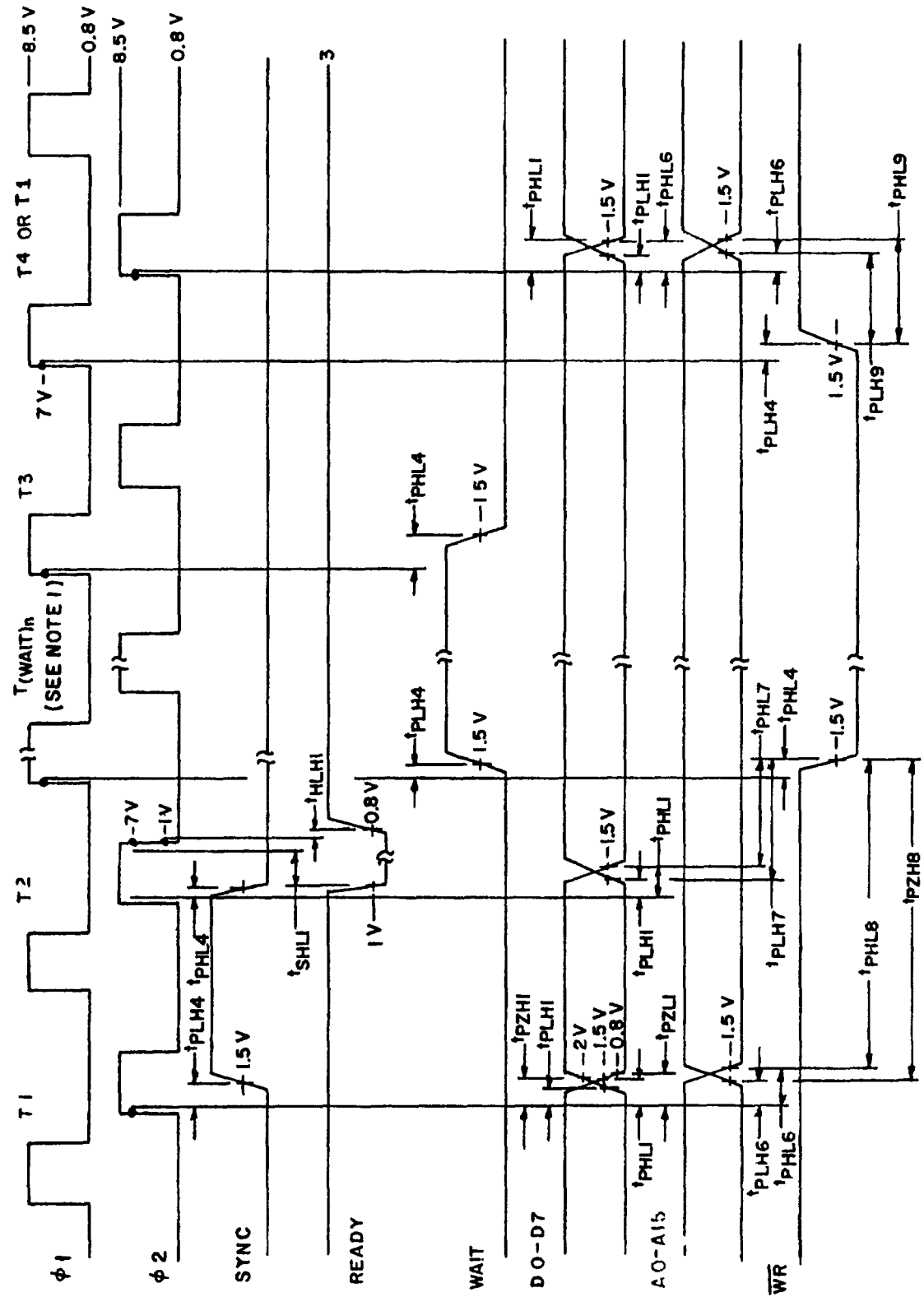
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NOTES:
1. It is not intended that the wait state be entered for every machine (m) cycle. When not performed omit the T wait cycle and disregard the READY and WAIT signals.
2. Unknown for T4 cycle.
3. All voltage levels are ±1%.

FIGURE 17. Read cycle with single or multiple wait cycles.

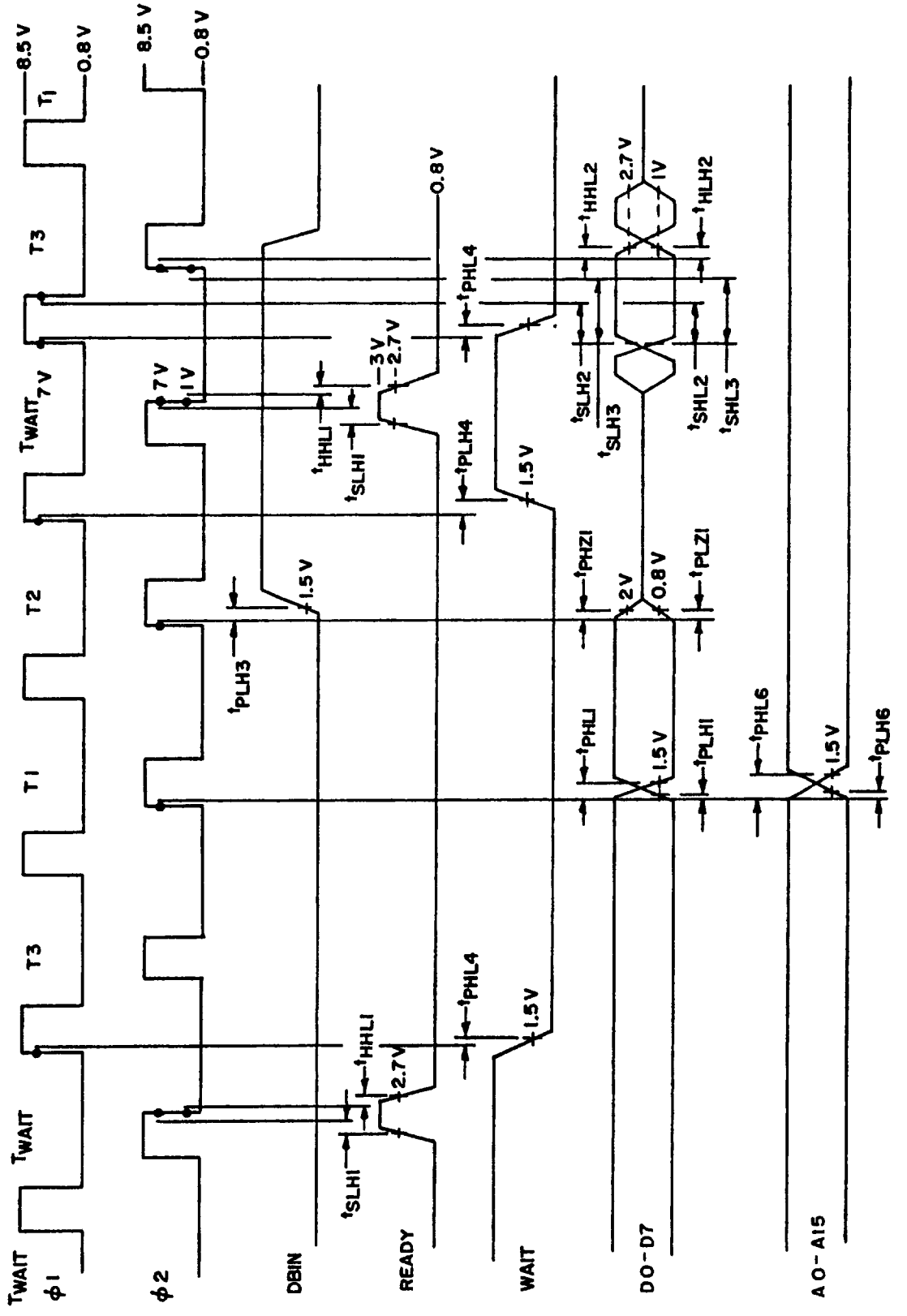
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NOTES:
1. It is not intended that the wait state be entered for every machine cycle (T1-T4) or T1-T3). When not performed, omit $T_{(WAIT)}$ cycle and disregard the READY and WAIT signals.
2. All voltage levels are ±1%.

FIGURE 18. Write cycle with single or multiple wait cycles.

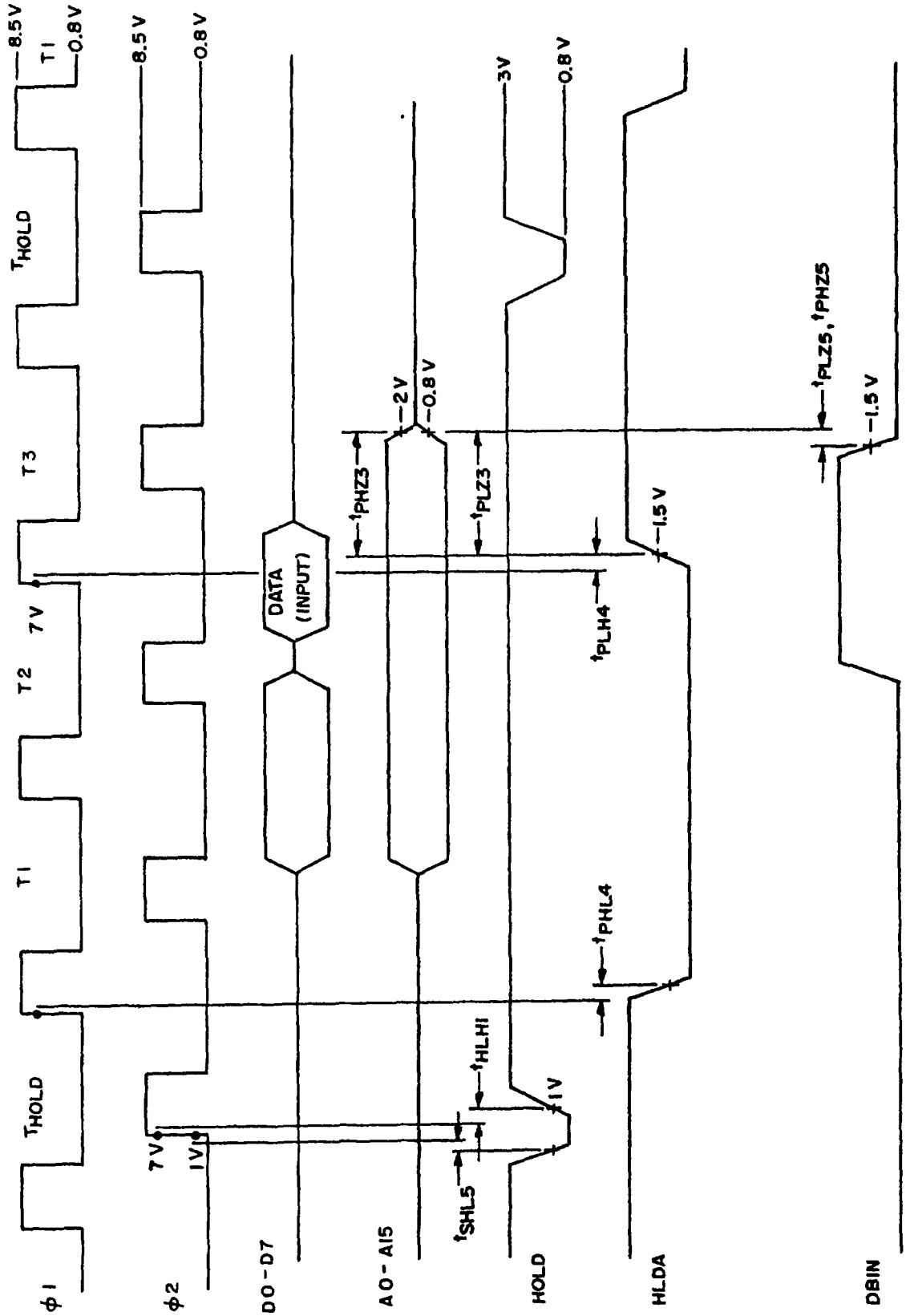
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NOTE: All voltage levels are ±1%.

FIGURE 19. Wait - machine cycle execute.

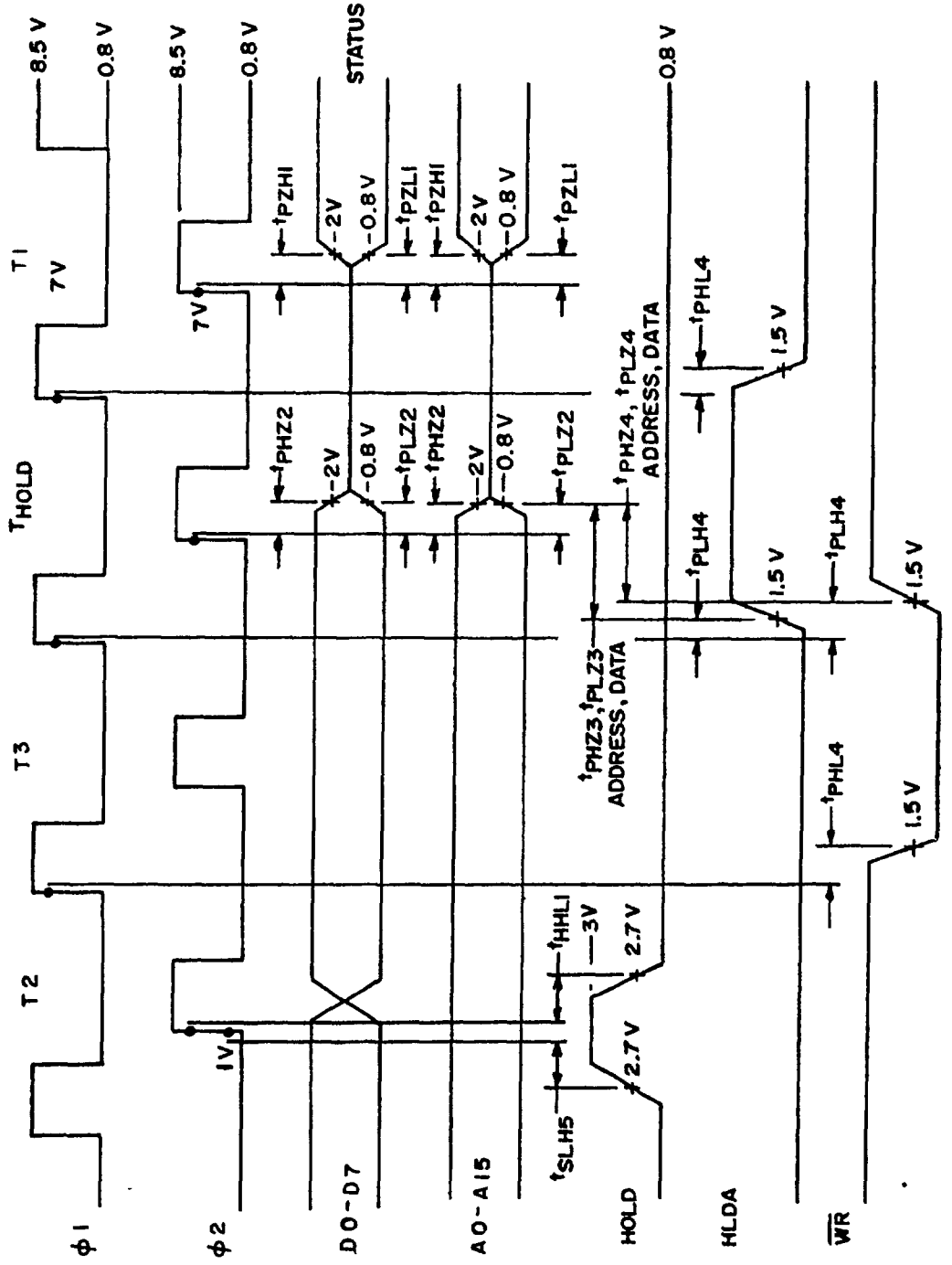
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NOTE: All voltage levels are ±1%.

FIGURE 20. Hold sequence - single machine cycle execute (read mode).

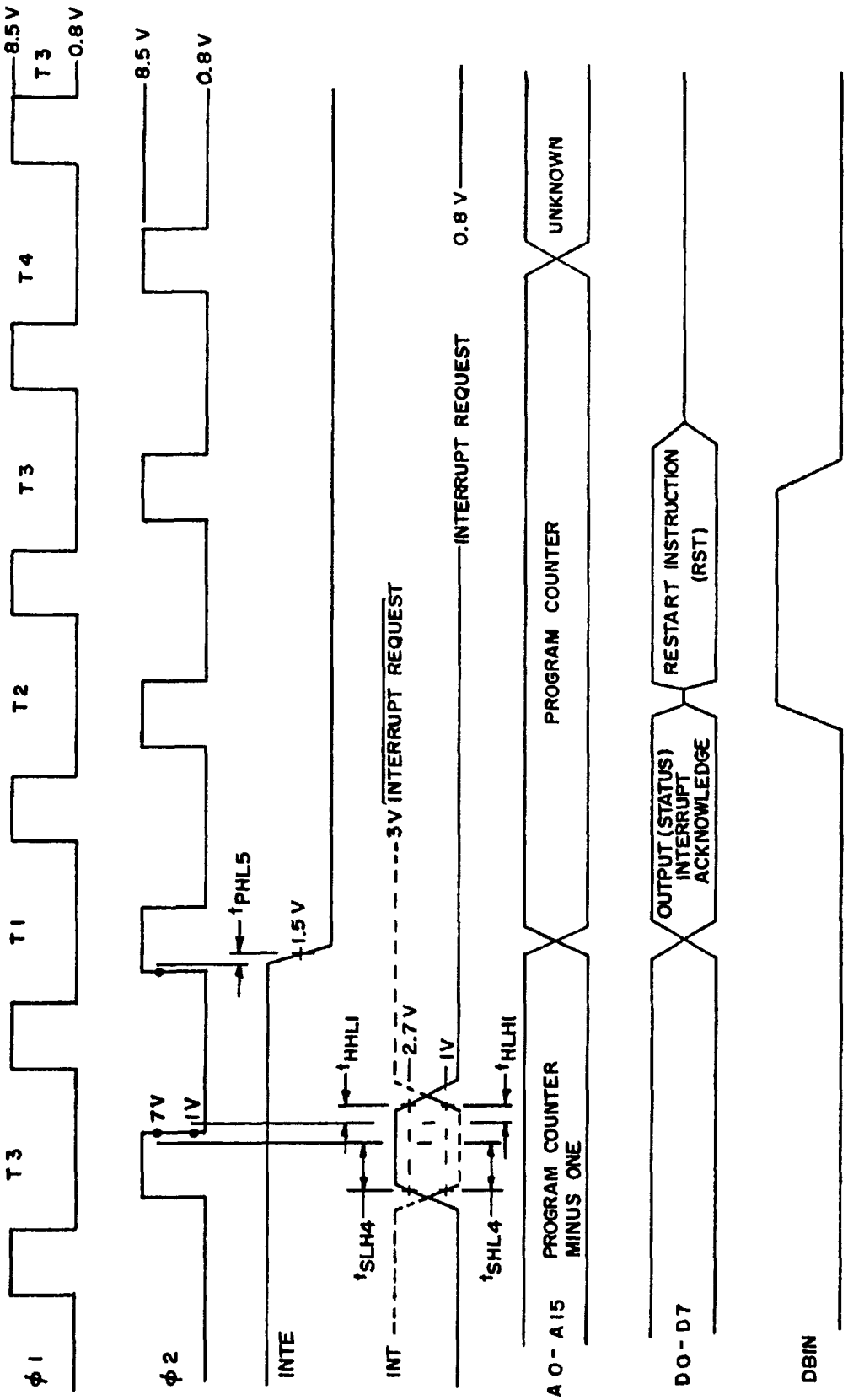
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NOTE: All voltage levels are ±1%.

FIGURE 21. Hold sequence - single cycle hold (write cycle).

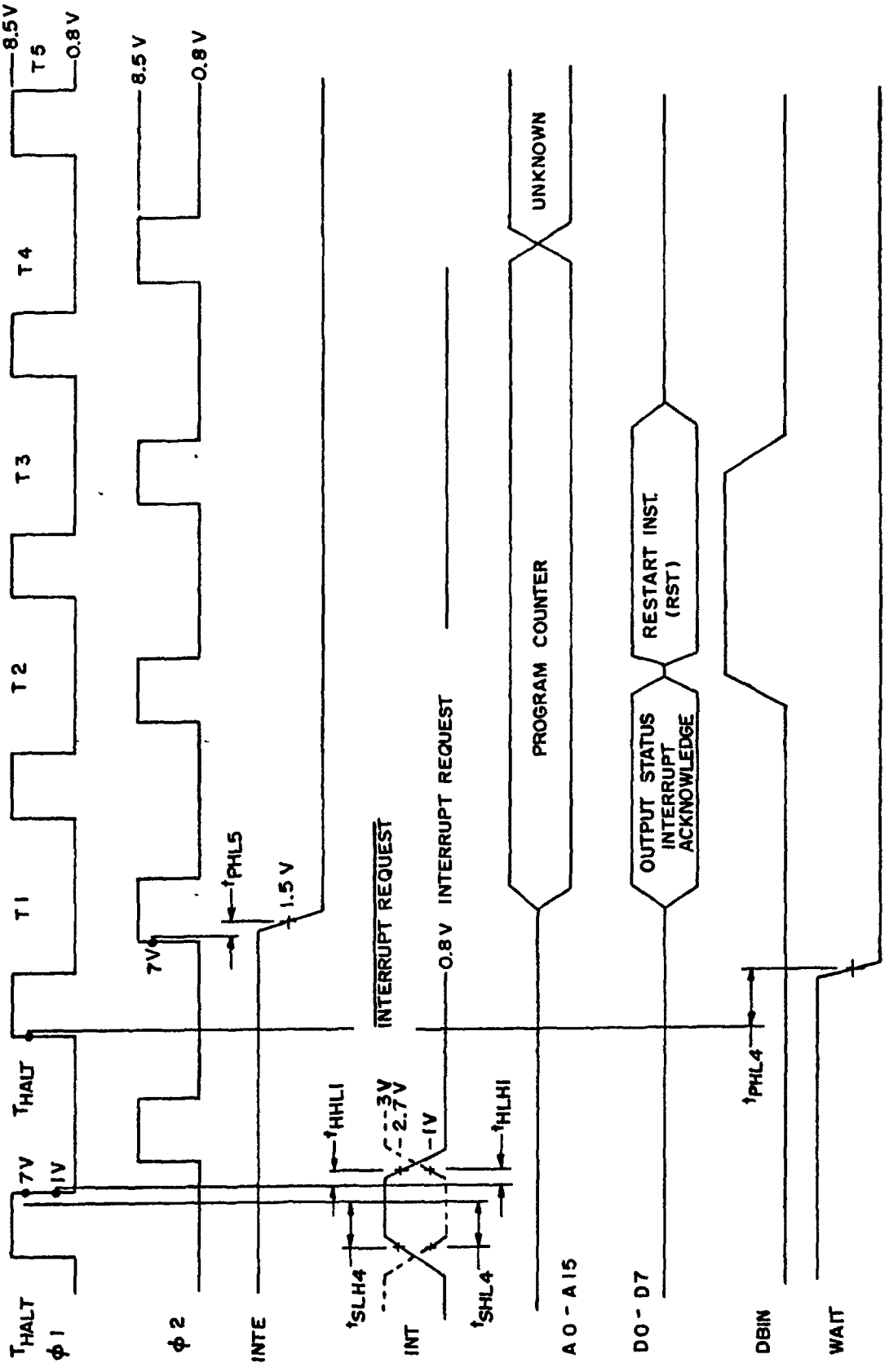
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NOTE: All voltage levels are ±1%.

FIGURE 22. Interrupt timing.

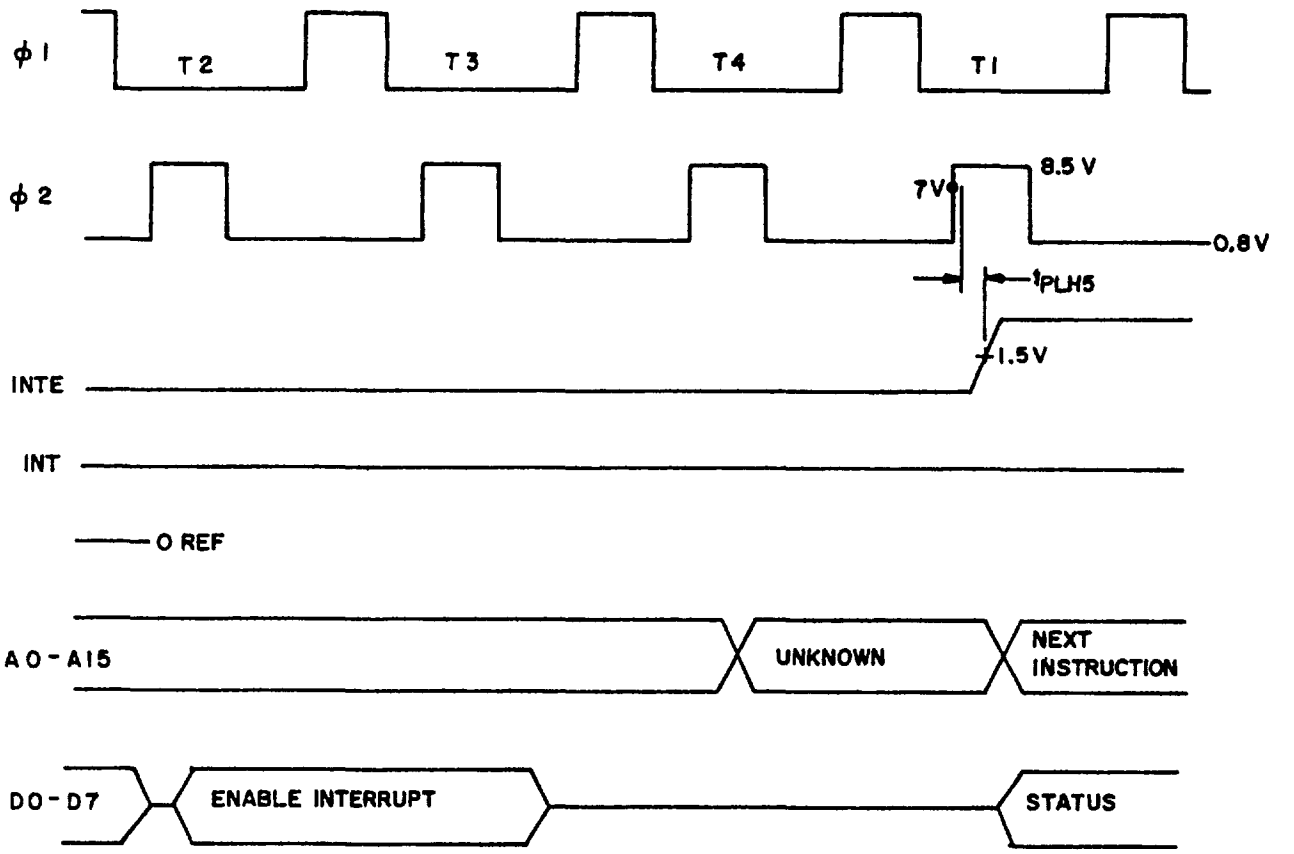
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NOTE: All voltage levels are ±1%.

FIGURE 23. Interrupt timing in halt mode.

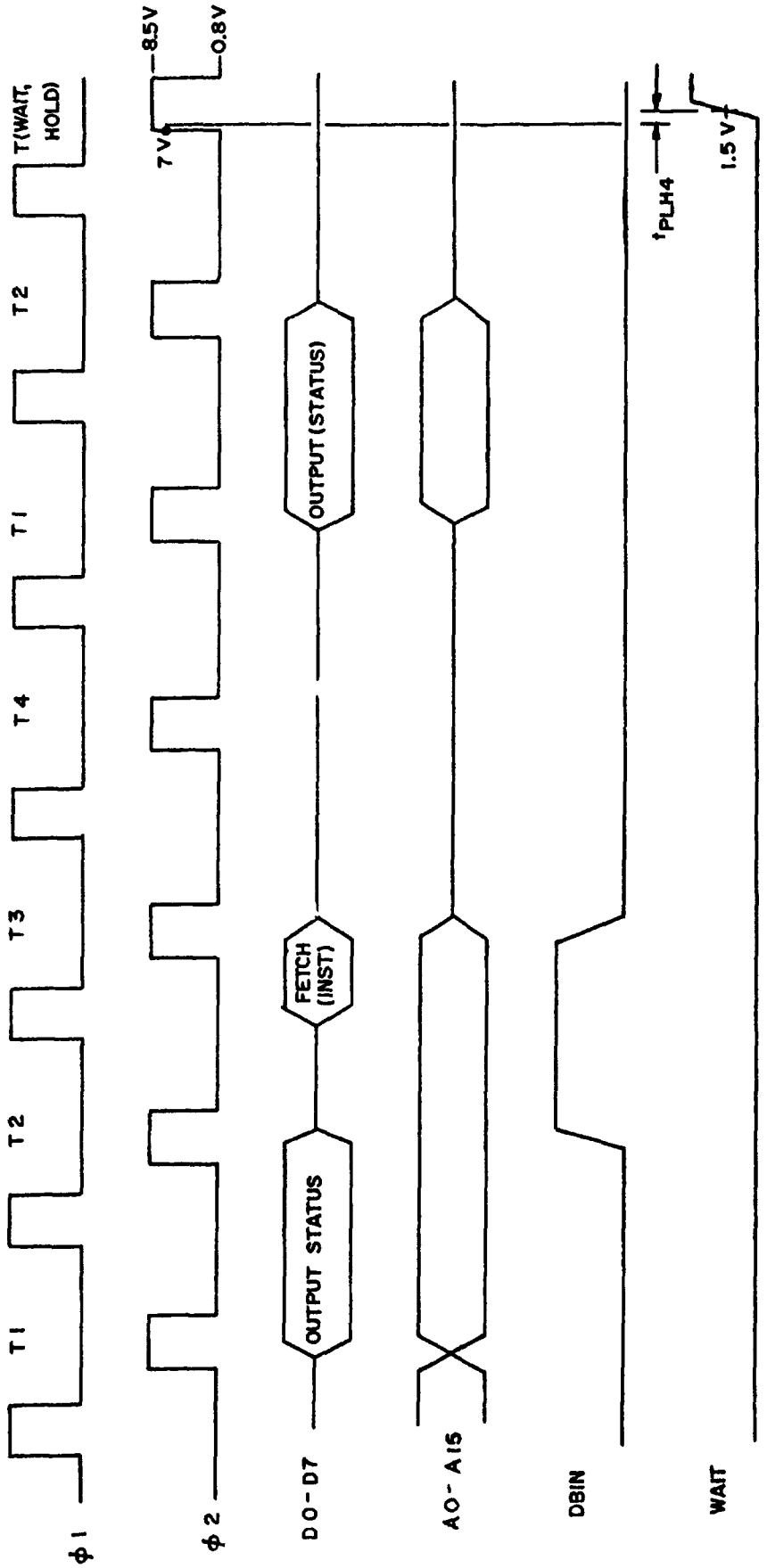
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NOTE: All voltage levels are ±1%.

FIGURE 24. Interrupt enable output - set INTE flip-flop.

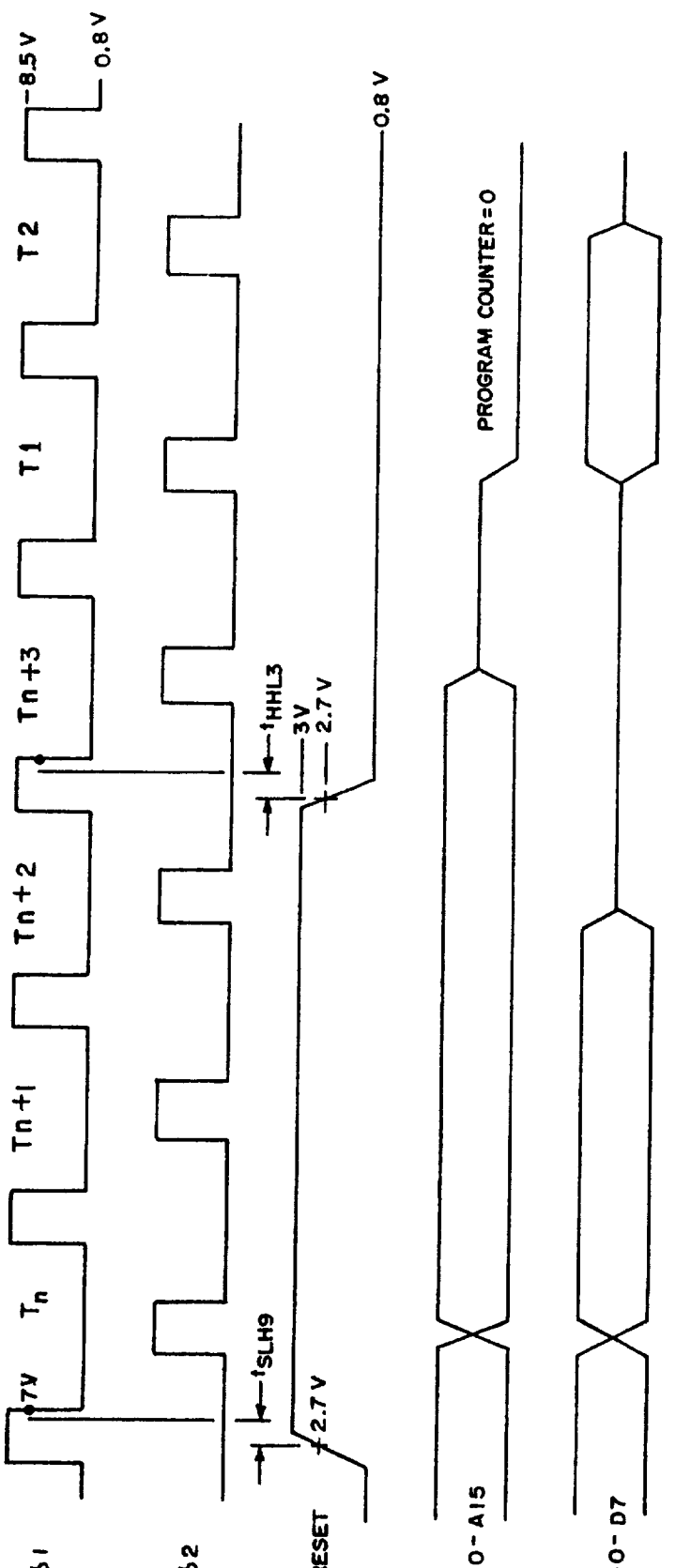
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NOTE: All voltage levels are ±1%.

FIGURE 25. Halt timing.

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NOTE: All voltage levels are ±1%.

FIGURE 26. Reset timing.

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TABLE XXV. Dynamic functional and propagation delay time test. 1/, 2/

Test condition	Test description	Input condition table	Output condition table	Waveform figures
M1	Dynamic functional and propagation delay time 3/ a	XXVI	XXX	16,17,18,19,20,21, 22,23,24,25,26
M2	Dynamic functional and propagation delay time 3/ b	XXVII	"	"
M3	Dynamic functional and propagation delay time 3/ c	XXVIII	"	"
M4	Dynamic functional and propagation delay time 3/ d	XXIX	"	"

1/ Load outputs as specified on figure 11.
2/ Connect power supplies in the following order: V_{BB}, V_{CC}, and V_{DD}.
3/ Test conditions M1 through M4:

a. Test condition M1 applies to the functional test vectors (see 6.5) and checks for logic outputs as detailed in 4/ c and for the following operating conditions and parameters:

Maximum frequency of operation.
Minimum 01 to 02 clock pulse delay.
Most negative power supply tolerances.
Minimum data bus transfer time (an internal propagation delay time check).
Worst case input setup and hold times and input threshold voltages.
Loaded output voltage levels.
Propagation delay time.

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b. Test condition M2 applies to the functional test vectors (see 6.5) and checks for logic outputs as detailed in 4/ c and for the following operating conditions and parameters:

Minimum operating frequency.
Minimum Ø1 to Ø2 clock pulse delay.
Most negative power supply tolerances.
Minimum data bus precharge time coupled with maximum data transfer time to the static registers.
Worst case input setup and hold times and input threshold voltages.
Loaded output voltage levels.
Propagation delay time.

c. Test condition M3 applies to the functional test vectors (see 6.5) and checks for logic outputs as detailed in 4/ c for the following operating conditions and parameters:

Minimum operating frequency.
Minimum Ø1 to Ø2 clock pulse delay.
Maximum Ø2 pulse width.
Worst case write conditions for dynamic accumulaltor.
Most negative power supply tolerances.
Worst case input setup and hold times and input threshold voltages.
Loaded output voltage levels.
Propagation delay time.

d. Test condition M4 applies to the functional test vectors (see 6.5) and checks for logic outputs as detailed in 4/ c and for the following operating conditions and parameters:
Maximum operating frequency.
Most positive power supply tolerances (worst case high-level threshold).
Minimum Ø1 pulse width.
Data setup to trailing edge of Ø1.
Worst case input setup and hold times and input threshold voltages.
Loaded output voltage levels.
Propagation delay time.

2/ The functional test vectors (see 6.5) shall be applied in the specified order with the following constraints:

a. Cycle definition. A cycle is the period beginning with a Ø1 leading edge and continuing until the next Ø1 leading edge. The input logic levels and expected output levels for a cycle (full or part of the cycle as defined below) are defined by one line (vector) (see 6.5).

b. Inputs. The basic cycle period is defined by Ø1. The Ø2 timing is referenced to Ø1. The other input timing conditions are in turn referenced to Ø1 or Ø2. Tables XXVI through XXIX specify the input voltage and timing conditions and also refer to the waveform figures which specify the waveform edges and voltage levels to which the timing parameters are referenced. Inputs are specified (see 6.5) by either A = V_{IH}, B = V_{IL}, or C = clock pulse. Clock inputs shall be driven by a positive pulse from V_{IL1} to V_{IH1} to V_{IL1} with the waveshape and timing as specified by the input timing tables and figure 16. The data bus is in the input mode whenever DBIN = 0 (see 6.5). The data bus inputs shall be driven by a return to complement waveform during these cycles as shown on figure 17. That is, the data bus inputs shall be at A (B) level from t_{SLH2} or t_{SLH3} (t_{SHL2} or t_{SHL3}) to t_{HHL2} (t_{HLH2}) and driven to

B (A) otherwise during the cycle. The intent of the HOLD (figures 20 and 21), INT (figures 22 and 23), and READY (figures 17, 18, and 19) are most easily met by also driving these signals with a return to complement waveform. RESET shall be a simple NRZ (nonreturn to zero) signal from t_{SLH9} to t_{HHL3} and t_{HHL3} to t_{SLH9}.

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c. Output levels and valid cycles. Outputs shall be monitored for logic level as follows:

H ≥ 3 V.
L ≤ 0.8 V.
Z ≤ 1.5 V and ≥1 V (output pin is in high-impedance mode and the level is determined by load circuit per figure 11).
X = don't care. No measurement is required, however, the pin(s) must not be driven by the test equipment during this cycle.

d. The vector set is organized in groups where each group is less than or equal to 1,024 vectors. This enables the use of this vector set on testers with a local memory greater than or equal to 1,024 locations. Each group of vectors must be applied in the order shown without pauses. The last vector of each group is as follows:

Group number	Last vector number
1	1,024
2	1,837
3	2,861
4	3,883
5	4,905
6	5,927
7	6,949
8	7,484
9	8,508
10	9,530
11	10,552
12	10,768
13	11,761

The groups must be run consecutively in the order shown. Between application of any two groups, the user has the option of:

* (1) No pause between the groups or vectors.

(2) Repeated application of the last vector of the group at test speed until the tester is ready with the next group. The outputs need not be monitored after the first application of the last vector in each group until the next group is ready.

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TABLE XXVI. Input conditions for test condition M1 of table XXV.

Parameter	Pins	Input 1/ condition	Unit	Figure
V_{BB}	20	-5.5 ±1%	V	---
V_{CC}	1	4.5 ±1%	"	---
V_{DD}	21	10.8 ±1%	"	---
V_{IH1}	All logic inputs except clocks	3 ±1%	"	---
V_{IH2}	2, 27	8.5 ±1%	"	---
V_{IL1}	All logic inputs including clocks	0.8 ±1%	"	---
t_{PERIOD}	2	480	ns	16
t_{01}	2	80	"	16
t_{02}	27	220	"	16
t_{SLH1}	3	120	"	19
t_{SHL1}	3	120	"	17, 18
t_{SLH2}	31, 32, 33, 34, 38, 37, 36, 35	130	"	17,19
t_{SHL2}	31, 32, 33, 34, 38, 37, 36, 35	130	"	17,19
t_{SLH3}	31, 32, 33, 34, 38, 37, 36, 35	130	"	17,19
t_{SHL3}	31, 32, 33, 34, 38, 37, 36, 35	130	"	17,19
t_{SLH4}	8	120	"	22, 23
t_{SHL4}	8	120	"	22, 23
t_{SLH5}	28	140	"	21
t_{SHL5}	28	140	"	20
t_{SLH6}	27	0	"	16
t_{SLH7}	27	80	"	16
t_{SHL8}	27	180	"	16
t_{SLH9}	29	75	"	26
t_{HLH1}	3, 8, 28	0	"	17,18,20, 22,23
t_{HHL1}	3, 8, 28	0	"	19,21,22, 23
t_{HLH2}	31, 32, 33, 34, 38, 37, 36, 35	50	"	17,19
t_{HHL2}	31, 32, 33, 34, 38, 37, 36, 35	50	"	17,19
t_{HHL3}	29	75	"	26

1/ All input timing parameters are ±2 ns.

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TABLE XXVII. Input conditions for test condition M2 of table XXV.

Parameter	Pins	Input 1/ condition	Unit	Figure
V_{BB}	20	-5.5 ±1%	V	---
V_{CC}	1	4.5 ±1%	"	---
V_{DD}	21	10.8 ±1%	"	---
V_{IH1}	All logic inputs except clocks	3 ±1%	"	---
V_{IH2}	2, 27	8.5 ±1%	"	---
V_{IL1}	All logic inputs including clocks	0.8 ±1%	"	---
t_{PERIOD}	2	2000	ns	16
t_{01}	2	80	"	16
t_{02}	27	220	"	16
t_{SLH1}	3	120	"	19
t_{SHL1}	3	120	"	17,18
t_{SLH2}	31, 32, 33, 34, 38, 37, 36, 35	130	"	17,19
t_{SHL2}	31, 32, 33, 34, 38, 37, 36, 35	130	"	17,19
t_{SLH3}	31, 32, 33, 34, 38, 37, 36, 35	130	"	17,19
t_{SHL3}	31, 32, 33, 34, 38, 37, 36, 35	130	"	17,19
t_{SLH4}	8	120	"	22,23
t_{SHL4}	8	120	"	22,23
t_{SLH5}	28	140	"	21
t_{SHL5}	28	140	"	20
t_{SLH6}	27	0	"	16
t_{SLH7}	27	80	"	16
t_{SHL8}	27	1,700	"	16
t_{SLH9}	29	75	"	26
t_{HLH1}	3, 8, 28	0	"	17,18,20, 22,23
t_{HHL1}	3, 8, 28	0	"	19,21,22, 23
t_{HLH2}	31, 32, 33, 34, 38, 37, 36, 35	50	"	17,19
t_{HHL2}	31, 32, 33, 34, 38, 37, 36, 35	50	"	17,19
t_{HHL3}	29	75	"	26

1/ All input timing parameters are ±2 ns.

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TABLE XXVIII. Input conditions for test condition M3 of table XXV.

Parameter	Pins	Input 1/ condition	Unit	Figure
V_{BB}	20	-5.5 ±1%	V	---
V_{CC}	1	4.5 ±1%	"	---
V_{DD}	21	10.8 ±1%	"	---
V_{IH1}	All logic inputs except clocks	3 ±1%	"	---
V_{IH2}	2, 27	8.5 ±1%	"	---
V_{IL1}	All logic inputs including clocks	0.8 ±1%	"	---
t_{PERIOD}	2	2000	ns	16
t_{O1}	2	80	"	16
t_{O2}	27	1840	"	16
t_{SLH1}	3	120	"	19
t_{SHL1}	3	120	"	17,18
t_{SLH2}	31, 32, 33, 34, 38, 37, 36, 35	130	"	17,19
t_{SHL2}	31, 32, 33, 34, 38, 37, 36, 35	130	"	17,19
t_{SLH3}	31, 32, 33, 34, 38, 37, 36, 35	130	"	17,19
t_{SHL3}	31, 32, 33, 34, 38, 37, 36, 35	130	"	17,19
t_{SLH4}	8	120	"	22,23
t_{SHL4}	8	120	"	22,23
t_{SLH5}	28	140	"	21
t_{SHL5}	28	140	"	20
t_{SLH6}	27	0	"	16
t_{SLH7}	27	80	"	16
t_{SHL8}	27	80	"	16
t_{SLH9}	29	75	"	26
t_{HLH1}	3, 8, 28	0	"	17,18,20, 22,23
t_{HHL1}	3, 8, 28	0	"	19,21,22, 23
t_{HLH2}	31, 32, 33, 34, 38, 37, 36, 35	50	"	17,19
t_{HHL2}	31, 32, 33, 34, 38, 37, 36, 35	50	"	17,19
t_{HHL3}	29	75	"	26

1/ All input timing parameters are ±2 ns.

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TABLE XXIX. Input conditions for test condition M4 of table XXV.

Parameter	Pins	Input 1/ condition	Unit	Figure
V_{BB}	20	-4.5 ±1%	V	---
V_{CC}	1	5.5 ±1%	"	---
V_{DD}	21	13.2 ±1%	"	---
V_{IH1}	All logic inputs except clocks	3 ±1%	"	---
V_{IH2}	2, 27	8.5 ±1%	"	---
V_{IL1}	All logic inputs including clocks	0.8 ±1%	"	---
t_{PERIOD}	2	500	ns	16
t_{O1}	2	60	"	16
t_{O2}	27	220	"	16
t_{SLH1}	3	120	"	19
t_{SHL1}	3	120	"	17,18
t_{SLH2}	31, 32, 33, 34, 36, 37, 36, 35	130	"	17,19
t_{SHL2}	31, 32, 33, 34, 38, 37, 36, 35	130	"	17,19
t_{SLH3}	31, 32, 33, 34, 38, 37, 36, 35	150	"	17,19
t_{SHL3}	31, 32, 33, 34, 38, 37, 36, 35	150	"	17,19
t_{SLH4}	8	120	"	22,23
t_{SHL4}	8	120	"	22,23
t_{SLH5}	28	140	"	21
t_{SHL5}	28	140	"	20
t_{SLH6}	27	120	"	16
t_{SLH7}	27	180	"	16
t_{SHL8}	27	80	"	16
t_{SLH9}	29	75	"	26
t_{HLH1}	3, 8, 28	0	"	17,18,20, 22,23
t_{HHL1}	3, 8, 28	0	"	19,21,22, 23
t_{HLH2}	31, 32, 33, 34, 38, 37, 36, 35	50	"	17,19
t_{HHL2}	31, 32, 33, 34, 38, 37, 36, 35	50	"	17,19
t_{HHL3}	29	75	"	26

1/ All input timing parameters are ±2 ns.

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TABLE XXX. Output conditions for test conditions M1-M4 of table XXV.

Parameter	Pins	Output cond 6/		Unit	Figure
		Min	Max		
V_{OH}	All outputs	3.7	---	V	---
V_{OL}	All outputs	---	0.45	V	---
t_{PLH1}	31,32,33,34,38,37,36,35	---	220	ns	17,18,19
t_{PHL1}	"	---	220	"	17,18,19
t_{PZL1}	"	---	220	"	17,18,21
t_{PZH1}	"	---	220	"	17,18,21
t_{PLZ1}	"	---	140	"	17,19
t_{PHZ1}	"	---	140	"	17,19
t_{PLZ2}	31,32,33,34,38,37,36,35,5,6,7,9,30, 11,12,13,14,15,39,40,17,18,19,16	---	130	"	21
t_{PHZ2}	31,32,33,34,38,37,36,35,5,6,7,9,30, 11,12,13,14,15,39,40,17,18,19,16	---	130	"	21
t_{PLH3}	25	25	150	"	17,19
t_{PHL3}	25	25	150	"	17
t_{PLZ3}	31,32,33,34,38,37,36,35,5,6,7,9,30, 11,12,13,14,15,39,40,17,18,19,16	1/	---	"	20,21
t_{PHZ3}	31,32,33,34,38,37,36,35,5,6,7,9,30, 11,12,13,14,15,39,40,17,18,19,16	1/	---	"	20,21
t_{PLH4}	23,24,22,4	---	140	"	17,18,19, 20,21,25
t_{PHL4}	23,24,22,4	---	140	"	17,18,19, 20,21,23
t_{PLZ4}	31,32,33,34,38,37,36,35,5,6,7,9,30, 11,12,13,14,15,39,40,17,18,19,16	2/	---	"	21
t_{PHZ4}	31,32,33,34,38,37,36,35,5,6,7,9,30, 11,12,13,14,15,39,40,17,18,19,16	2/	---	"	21
t_{PLH5}	26	---	200	"	24
t_{PHL5}	26	---	200	"	22,23
t_{PLZ5}	5,6,7,9,30,11,12,13,14,15,39,40,17, 18,19,16	-20	---	"	20
t_{PHZ5}	"	-20	---	"	20
t_{PLH6}	"	---	200	"	17,18,19
t_{PHL6}	"	---	200	"	17,18,19

See footnotes at end of table.

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TABLE XXX. Output conditions for test conditions M1-M4 of table XXV - Continued.

Parameter	Pins	Output cond 6/		Unit	Figure
		Min	Max		
t_{PLH7}	31,32,33,34,38,37,36,35	3/	---	ns	18
t_{PHL7}	31,32,33,34,38,37,36,35	3/	---	"	"
t_{PZH8}	5,6,7,9,30,11,12,13,14,15,39,40,17,18,19,16	4/	---	"	"
t_{PHL8}	5,6,7,9,30,11,12,13,14,15,39,40,17,18,19,16	4/	---	"	"
t_{PLH9}	31,32,33,34,38,37,36,35,5,6,7,9,30,11,12,13,14,15,39,40,17,18,19,16	5/	---	"	"
t_{PHL9}	31,32,33,34,38,37,36,35,5,6,7,9,30,11,12,13,14,15,39,40,17,18,19,16	5/	---	"	"

1/ Minimum t_{PLZ3} and t_{PHZ3} = t_{SLH7} + t_{TLHO2} - 50 ns (t_{TLHO2} = transition time of O2 from 1 to 7 V-refer to figure 16).

2/ Minimum t_{PLZ4} and t_{PHZ4} = t_{SLH7} + t_{TLHO2} - 10 ns.

3/ Minimum t_{PLH7} and t_{PHL7} = t_{PERIOD} - t_{SLH7} - t_{TLHO2} - 170 ns.

4/ Minimum t_{PZH8} and t_{PHL8} = 2 X t_{PERIOD} - t_{SLH7} - t_{TLHO2} - 140 ns.

5/ Minimum t_{PLH9} and t_{PHL9} = t_{SLH7} + t_{TLHO2} + 10 ns.

6/ All outputs will be checked at their maximum delay time for a period of 10 ns.

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TABLE XXXI. Output transition time (t_{TLH}) test.

Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
N1	D0	Output transition time (t_{TLH})	31	3	35	ns	27,28
N2	D1	"	32	"	"	"	"
N3	D2	"	33	"	"	"	"
N4	D3	"	34	"	"	"	"
N5	D4	"	38	"	"	"	"
N6	D5	"	37	"	"	"	"
N7	D6	"	36	"	"	"	"
N8	D7	"	35	"	"	"	"
N9	A0	"	5	"	"	"	"
N10	A1	"	6	"	"	"	"
N11	A2	"	7	"	"	"	"
N12	A3	"	9	"	"	"	"
N13	A4	"	30	"	"	"	"
N14	A5	"	11	"	"	"	"
N15	A6	"	12	"	"	"	"
N16	A7	"	13	"	"	"	"
N17	A8	"	14	"	"	"	"
N18	A9	"	15	"	"	"	"
N19	A10	"	39	"	"	"	"
N20	A11	"	40	"	"	"	"

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TABLE XXXI. Output transition time (t_{TLH}) test - Continued.

Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
N21	A12	Output transition time (t_{TLH})	17	3	35	ns	27,28
N22	A13	"	18	"	"	"	"
N23	A14	"	19	"	"	"	"
N24	A15	"	16	"	"	"	"
N25	SYNC	"	23	"	"	"	"
N26	HLDA	"	22	"	"	"	"
N27	INTE	"	26	"	"	"	"
N28	WAIT	"	4	"	"	"	"
N29	DBIN	"	25	"	"	"	"
N30	$\overline{WR}$	"	24	"	"	"	"

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TABLE XXXII. Output transition time (t_{THL}) test.

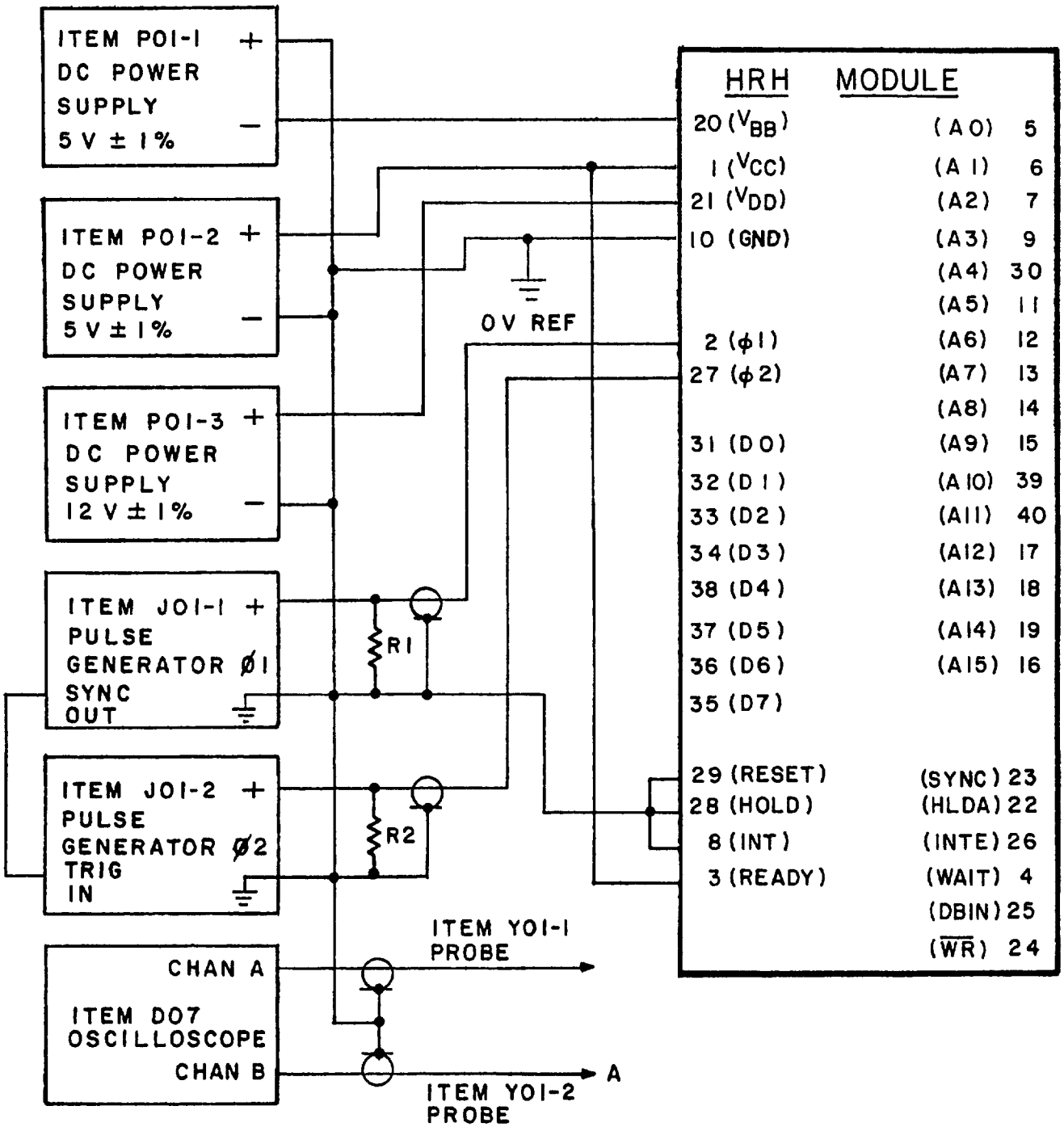
Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
P1	D0	Output transition time (t_{THL})	31	2	25	ns	27,28
P2	D1	"	32	"	"	"	"
P3	D2	"	33	"	"	"	"
P4	D3	"	34	"	"	"	"
P5	D4	"	38	"	"	"	"
P6	D5	"	37	"	"	"	"
P7	D6	"	36	"	"	"	"
P8	D7	"	35	"	"	"	"
P9	A0	"	5	"	"	"	"
P10	A1	"	6	"	"	"	"
P11	A2	"	7	"	"	"	"
P12	A3	"	9	"	"	"	"
P13	A4	"	30	"	"	"	"
P14	A5	"	11	"	"	"	"
P15	A6	"	12	"	"	"	"
P16	A7	"	13	"	"	"	"
P17	A8	"	14	"	"	"	"
P18	A9	"	15	"	"	"	"
P19	A10	"	39	"	"	"	"
P20	A11	"	40	"	"	"	"

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TABLE XXXII. Output transition time (t_{THL}) test - Continued.

Test condition	Parameter		Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
P21	A12	Output transition time (t_{THL})	17	2	25	ns	27,28
P22	A13	"	18	"	"	"	"
P23	A14	"	19	"	"	"	"
P24	A15	"	16	"	"	"	"
P25	SYNC	"	23	"	"	"	"
P26	HLDA	"	22	"	"	"	"
P27	INTE	"	26	"	"	"	"
P28	WAIT	"	4	"	"	"	"
P29	DBIN	"	25	"	"	"	"
P30	$\overline{WR}$	"	24	"	"	"	"

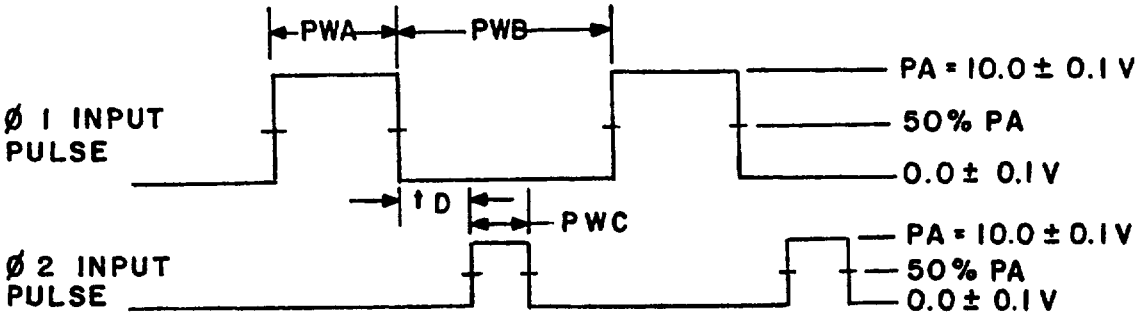
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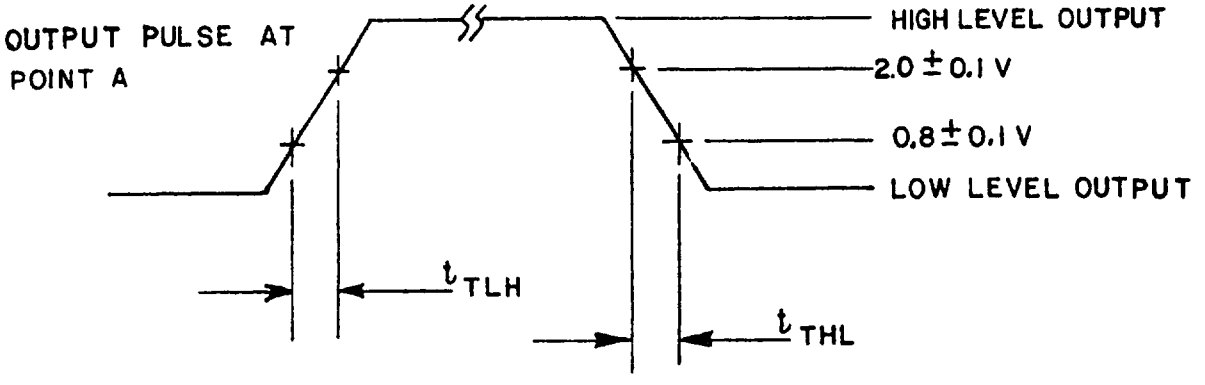


NOTES:
1. R1 = R2 = 51.1Ω ±1%, 2 W.
2. Load outputs as specified on figure 11.
3. Connect power supplies in the following order: V_{BB}, V_{CC}, and V_{DD}.

* FIGURE 27. Output transition time (t_{TLH} and t_{THL}) test circuit for table XXXI and XXXII.

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NOTES:
1. Input pulse parameters:
   PWA = 3 µs ±10%.
   PWB = 7 µs ±10%.
   PWC = 2 µs ±10%.
   t_D = 2.5 µs ±10%.
   Ø1: $t_{TLH} = t_{THL}$ = 20 ±20 ns maximum.
   Ø2: $t_{TLH} = t_{THL}$ = 20 ±20 ns maximum.
2. Before making the measurements, verify that the output high level is $\geq$ 3.7 V and output low level is $\leq$ 0.45 V.
3. All voltages are referenced to ground.

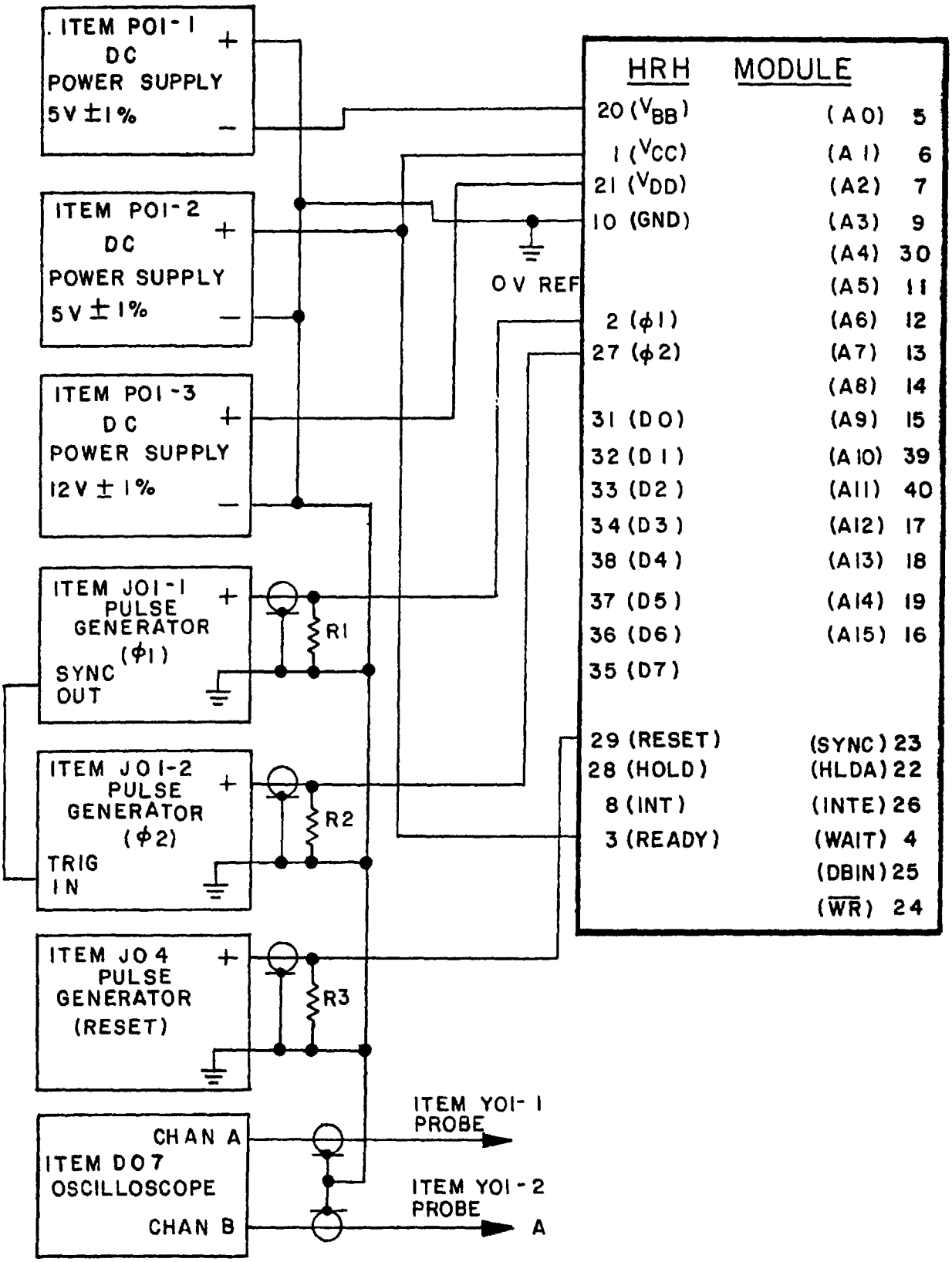
FIGURE 28. Output transition time (t_{TLH} and t_{THL}) test circuit waveform for tables XXXI and XXXII.

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TABLE XXXIII. Power supply transient amplitude test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
Q1	V_{BB} Power supply transient amplitude	20	0	400	mV	29,30
Q2	V_{CC} Power supply transient amplitude	1	0	400	mV	29,30
Q3	V_{DD} Power supply transient amplitude	21	0	400	mV	29,30

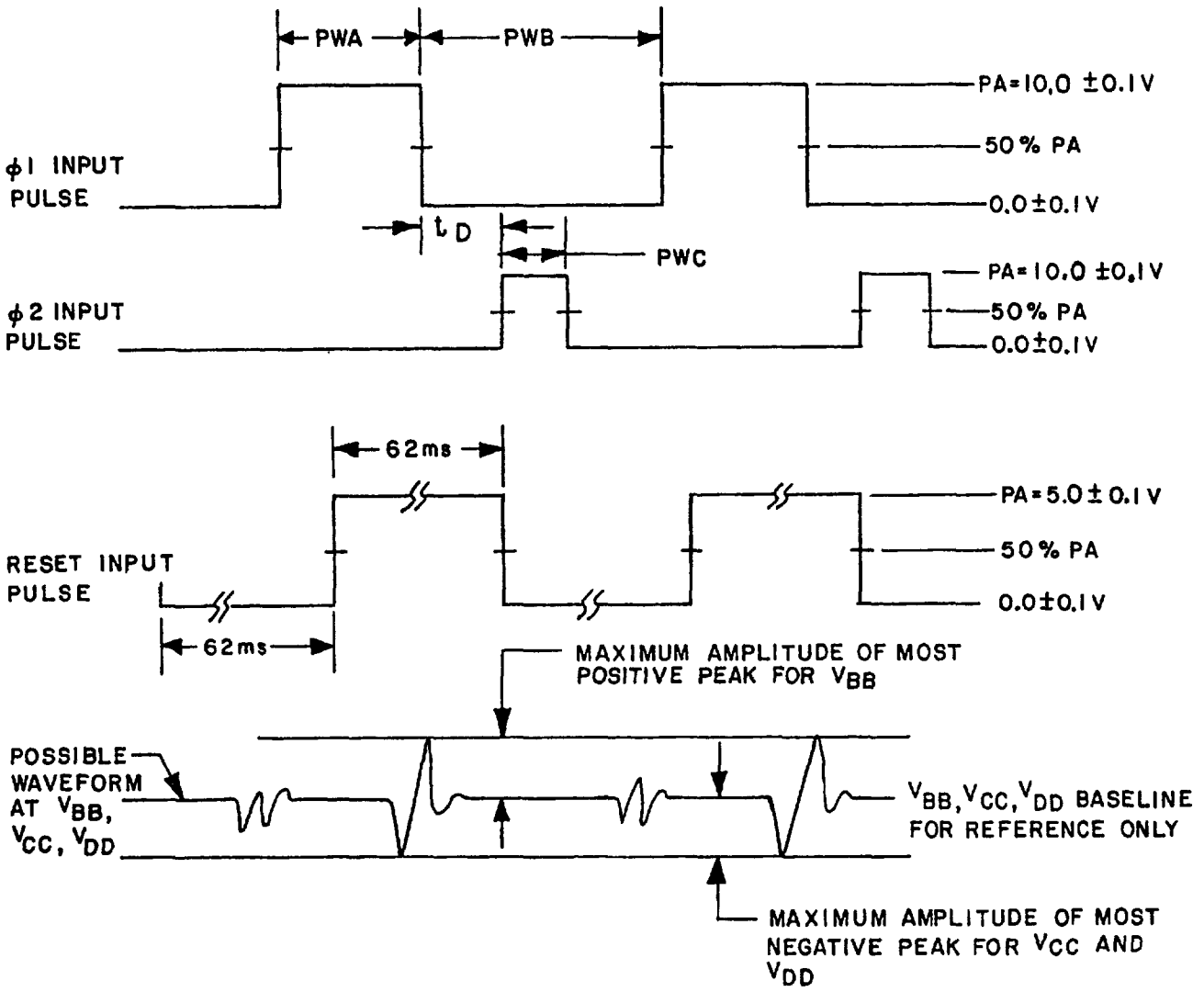
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NOTE: R1-R3 = 51.1Ω ±1%, 2 W.

FIGURE 29. Power supply transient amplitude test circuit for table XXXIII.

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NOTES:
1. Input pulse parameters:
   PWA = 3 µs ±10%.
   PWB = 7 µs ±10%.
   PWC = 2 µs ±10%.
   t_D = 2.5 µs ±10%.
   Ø1: $t_{TLH} = t_{THL}$ = 20 ±20 ns maximum.
   Ø2: $t_{TLH} = t_{THL}$ = 20 ±20 ns maximum.
2. Waveform transition times (t_{TLH} and t_{THL}) are measured from 10% PA to 80% PA points for clocks (Ø1 and Ø2) and for the RESET input. All measurements are with the device under test connected.

FIGURE 30. Power supply transient amplitude test circuit waveforms for table XXXIII.

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TABLE XXXIV. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
R1	V_{BB} Power supply filter capacitance	Open	20	4.06	7.54	µF	31
R2	V_{CC} Power supply filter capacitance	Open	1	4.06	7.54	µF	31
R3	V_{DD} Power supply filter capacitance	Open	21	4.06	7.54	µF	31

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TABLE XXXV. Input and output capacitance test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
S1	O1	Input capacitance	2	Open	2	25	pF	31
S2	O2	"	27	"	"	25	"	"
S3	D0	"	31	"	"	15	"	"
S4	D1	"	32	"	"	"	"	"
S5	D2	"	33	"	"	"	"	"
S6	D3	"	34	"	"	"	"	"
S7	D4	"	38	"	"	"	"	"
S8	D5	"	37	"	"	"	"	"
S9	D6	"	36	"	"	"	"	"
S10	D7	"	35	"	"	"	"	"
S11	RESET	"	29	"	"	"	"	"
S12	HOLD	"	28	"	"	"	"	"
S13	INT	"	8	"	"	"	"	"
S14	READY	"	3	"	"	"	"	"
T1	A0	Output capacitance	5	"	"	20	"	"
T2	A1	"	6	"	"	"	"	"
T3	A2	"	7	"	"	"	"	"
T4	A3	"	9	"	"	"	"	"

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TABLE XXXV. Input and output capacitance test - Continued.

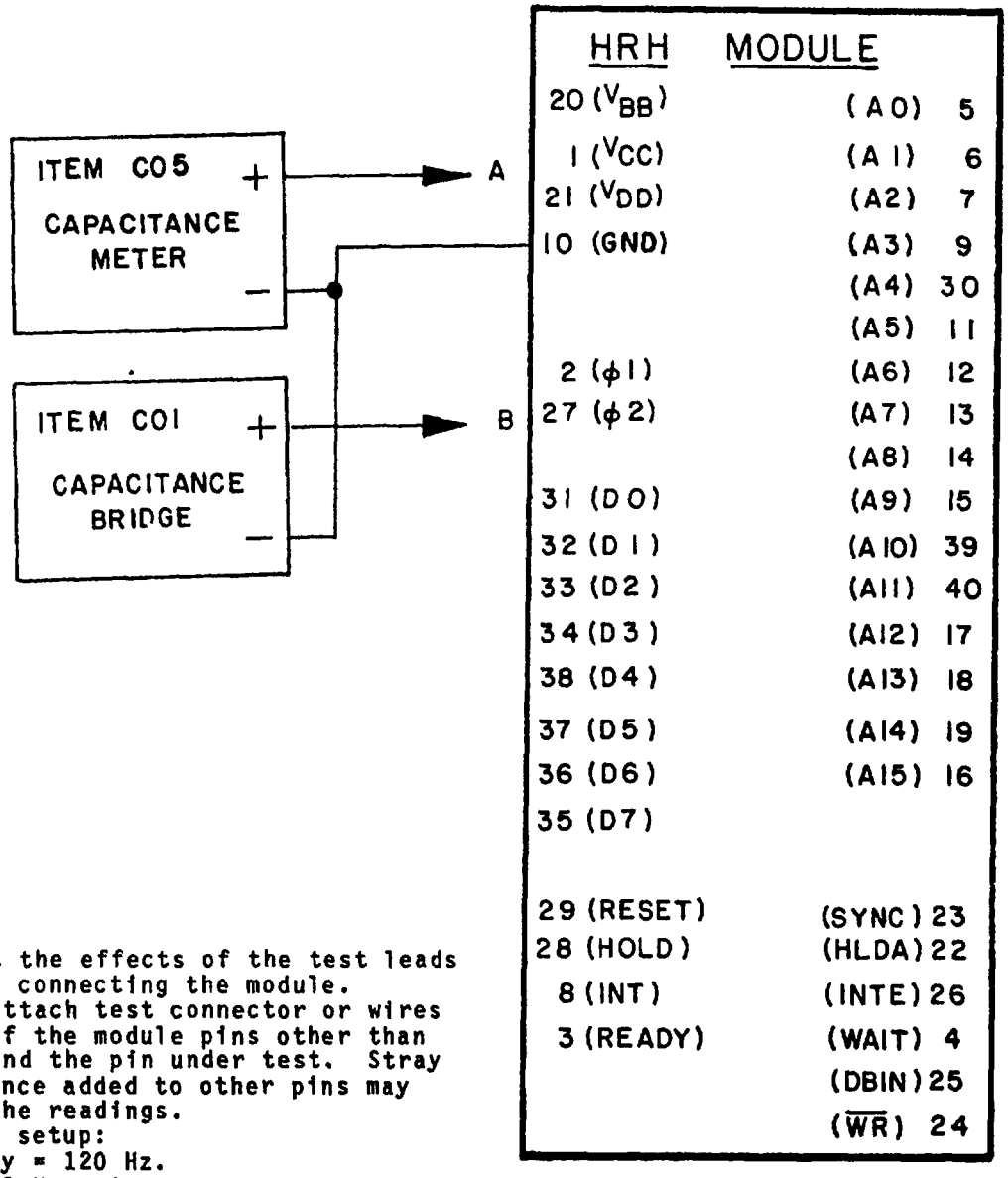
Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
T5	A4	Output capacitance	30	Open	2	20	pF	31
T6	A5	"	11	"	"	"	"	"
T7	A6	"	12	"	"	"	"	"
T8	A7	"	13	"	"	"	"	"
T9	A8	"	14	"	"	"	"	"
T10	A9	"	15	"	"	"	"	"
T11	A10	"	39	"	"	"	"	"
T12	A11	"	40	"	"	"	"	"
T13	A12	"	17	"	"	"	"	"
T14	A13	"	18	"	"	"	"	"
T15	A14	"	19	"	"	"	"	"
T16	A15	"	16	"	"	"	"	"
T17	SYNC	"	23	"	"	"	"	"
T18	HLDA	"	22	"	"	"	"	"

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TABLE XXXV. Input and output capacitance test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
T19	INTE	Output capacitance	26	Open	2	20	pF	31
T20	WAIT	"	4	"	"	"	"	"
T21	DBIN	"	25	"	"	"	"	"
T22	$\overline{WR}$	"	24	"	"	"	"	"

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Item C01 setup:
   Frequency = 120 Hz.
   V_{pk} = 0.3 V maximum.

* FIGURE 31. Power supply filter input, and output capacitance test circuit for tables XXXIV and XXXV.

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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